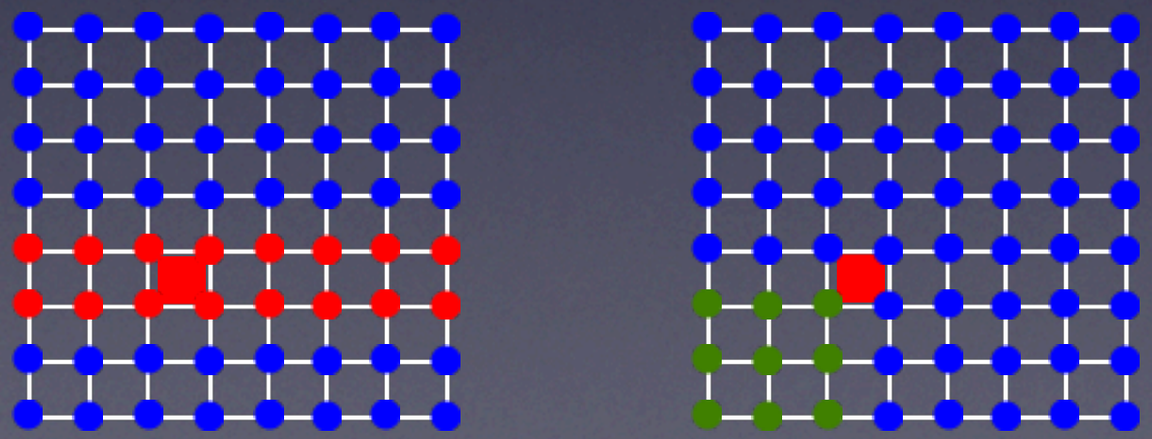
Punching Holes and Quantum Fault-Tolerance

Dave Bacon
Department of Computer Science & Engineering
Department of Physics
University of Washington

0+1

0+1



ENIAC, 1946

- 17,468 vacuum tubes
- 7,200 crystal diodes
- 1,500 relays
- 70,000 resistors
- 10,000 capacitors
- ~5,000,000 hand-soldered joints

~ one vacuum tube failure / day
(equivalent to ~ one transistor failure /second today)

Alternative History

Imagine: no transistors, no integrated circuits...



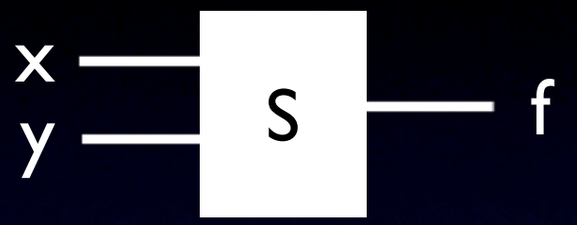




Could we build a computer out of unreliable components?

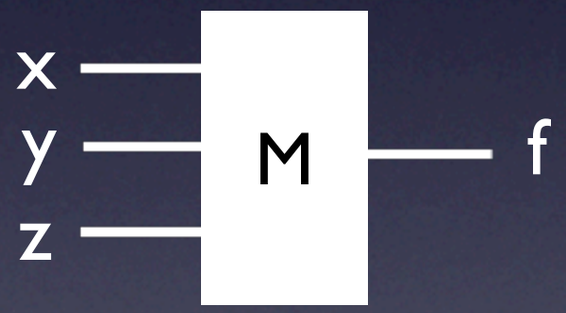
J. von Neumann, "Probabilistic Logics and the Synthesis of Reliable Organism from Unreliable Components" 1956

Von Neumann's Model



NAND gate

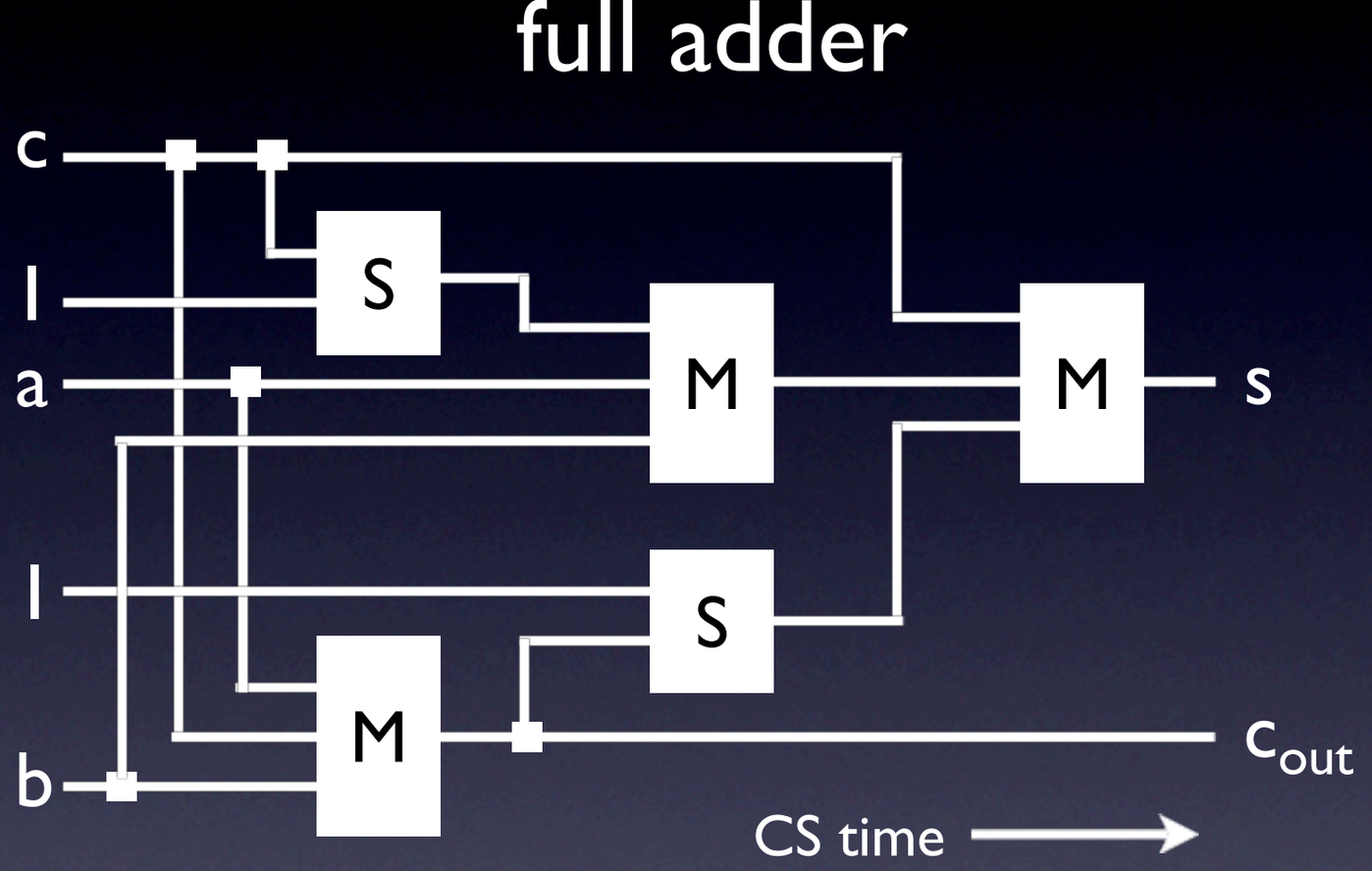
x	y	f
0	0	1
0	1	1
1	0	1
1	1	0



majority gate

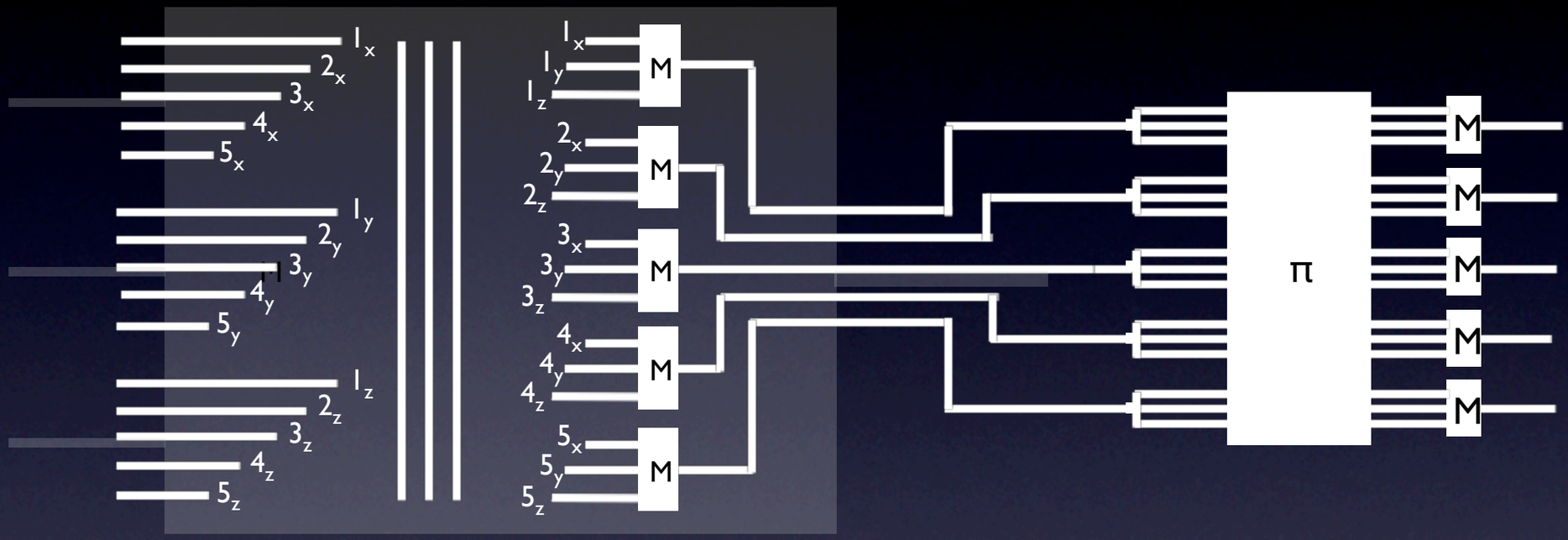
x	y	z	f
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

full adder



suppose each gate fails (outputs wrong answer) *independently* and with probability p

Von Neumann's Solution

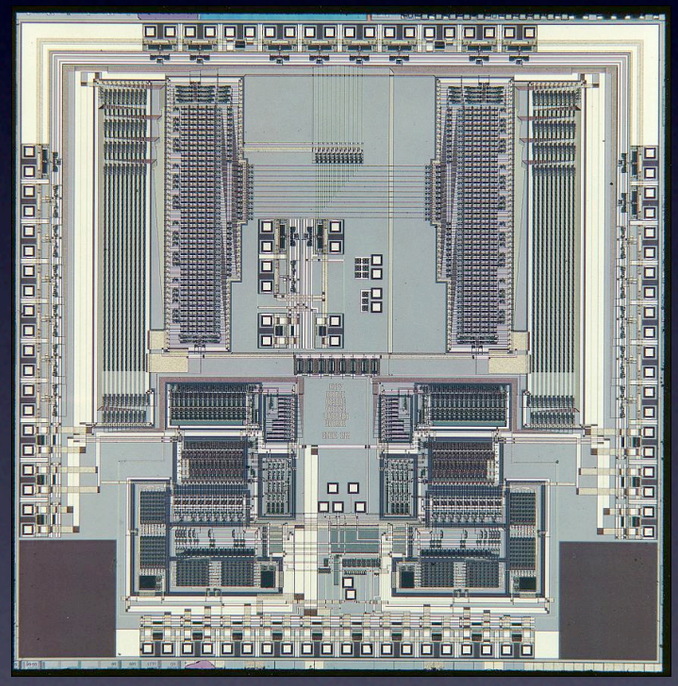


gate error correction

Theorem (von Neumann): A circuit of g gates can be made to fail with probability ε using $O(g \log (g/\varepsilon))$ gates which fail with probability p, assuming $p<p_{th}$ (vN estimated $p_{th} \sim 1\%$)

We Don't Need Your Stickin' Engineering

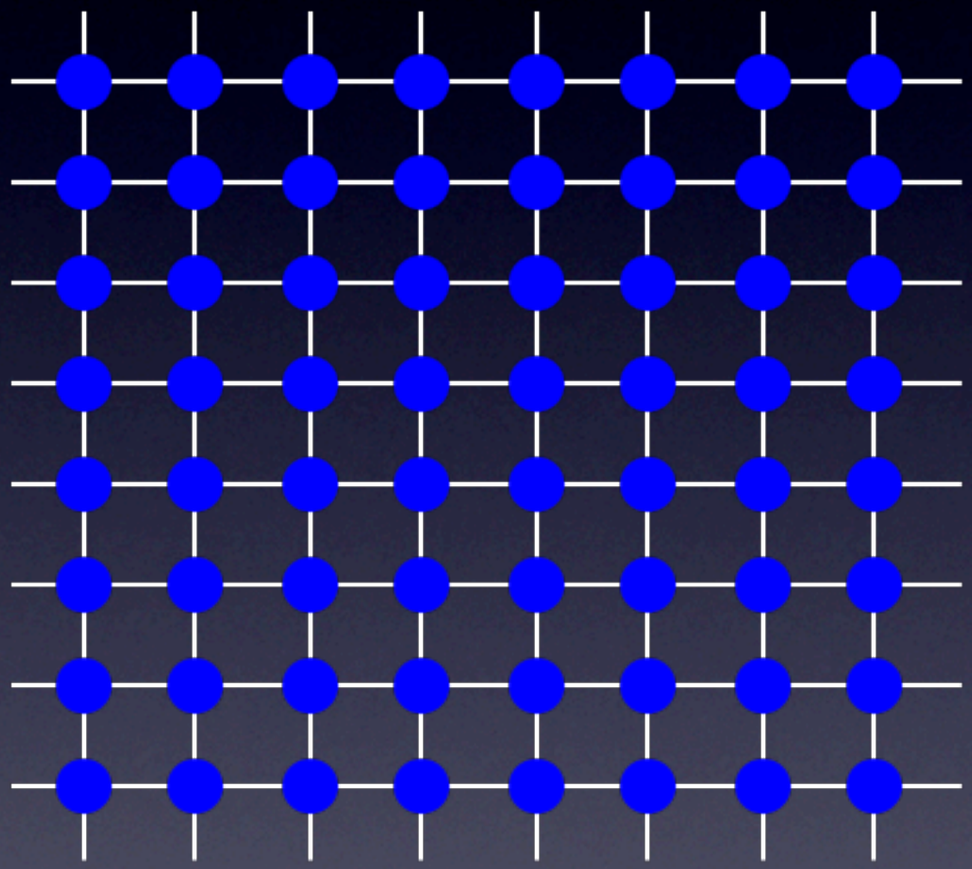






There are distinct *physical* reasons why robust classical computation is possible

Cartoon Guide to Classical Physical Fault-Tolerance



$+1$

-1

spin

$s_i \in \{+1, -1\}$

Energy

$$E = -J \sum_{\langle i,j \rangle} s_i s_j$$

neighbors

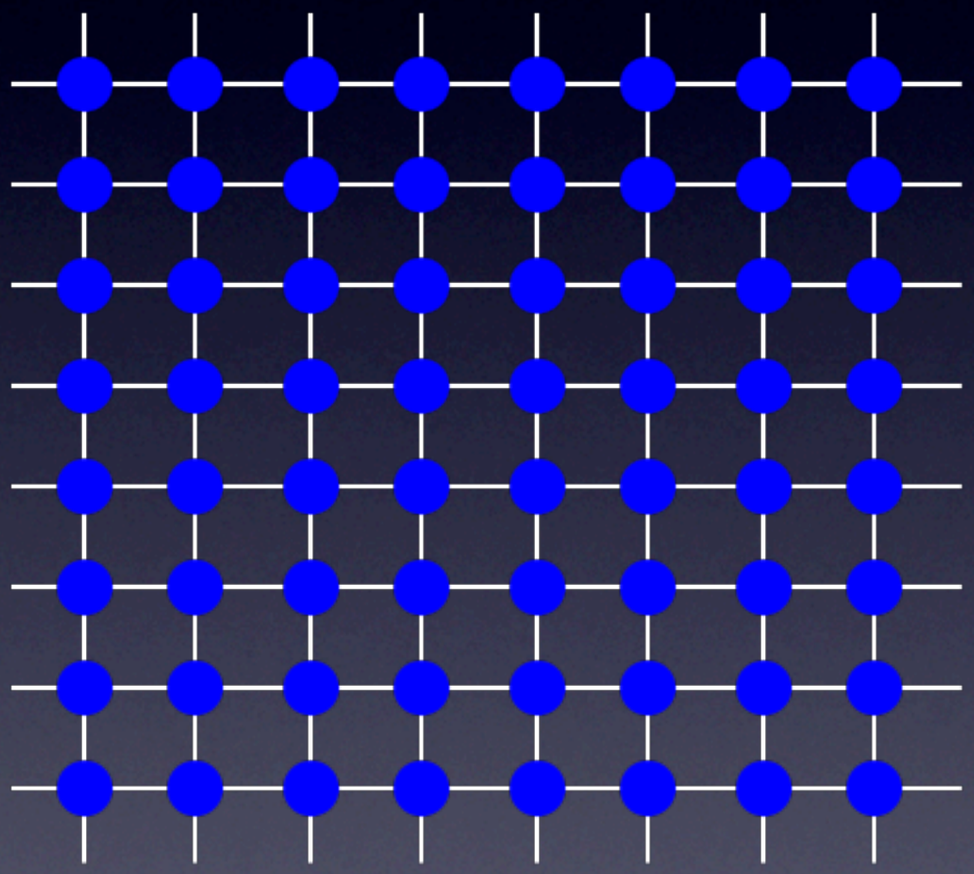
total magnetization

$$M = \sum_i s_i$$

"0" as $M > N_0$

"1" as $M < -N_0$

Cartoon Guide to Classical Physical Fault-Tolerance



"bare" noise time step

N_r

runs of 1-3

Simple noise model

1. Pick spin at random

2. If flipping spin would decrease energy

a. flip spin

3. If flipping spin would increase energy:

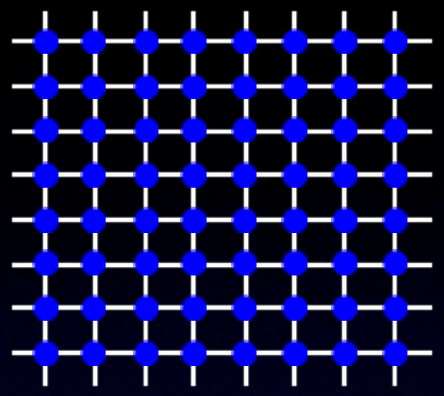
a. flip spin with probability

$$\exp(-\beta \Delta E) \quad \beta = T^{-1}$$

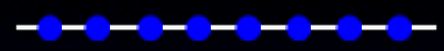
$\beta = T^{-1}$

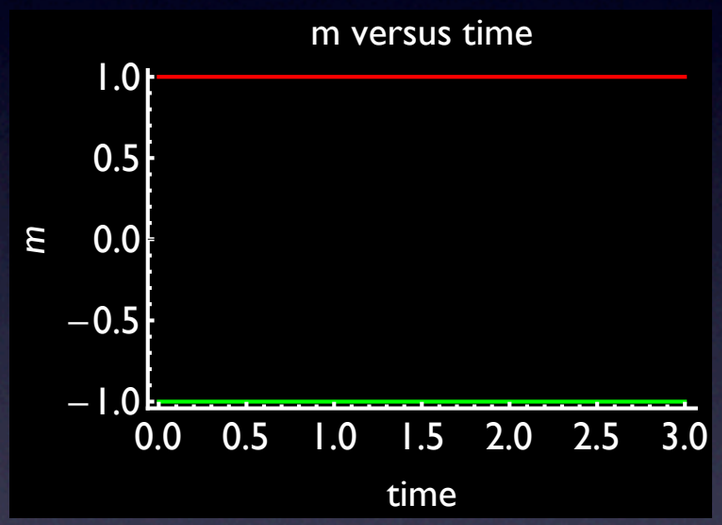
Memory

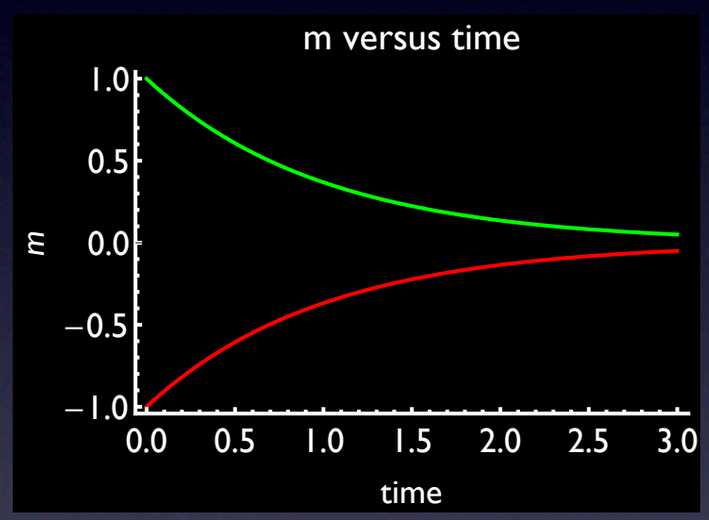
2D

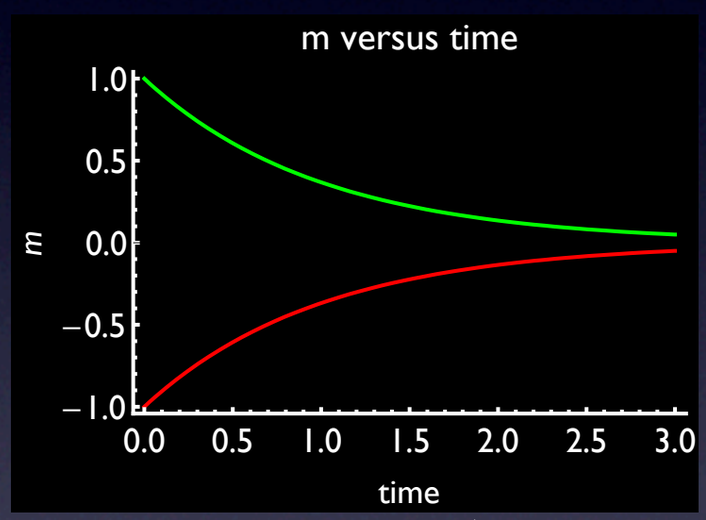


1D





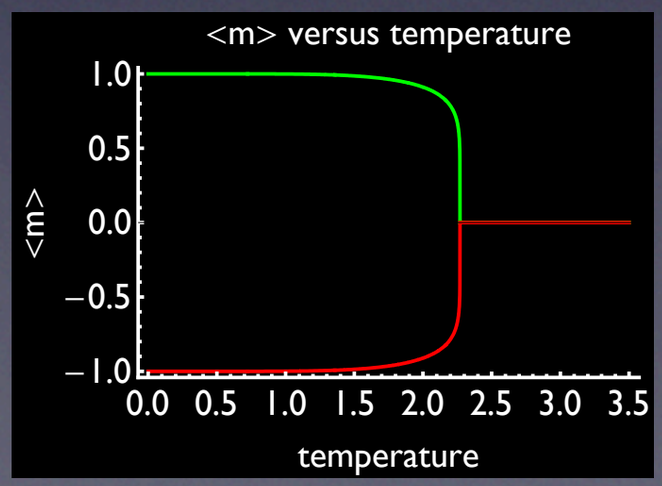


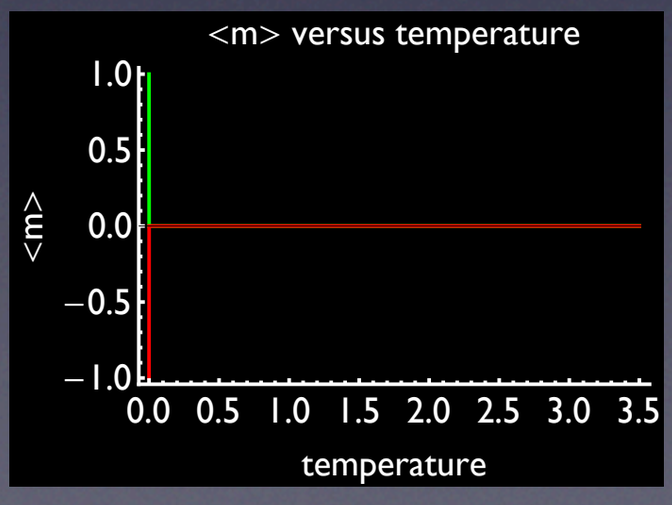


$T < T_C$

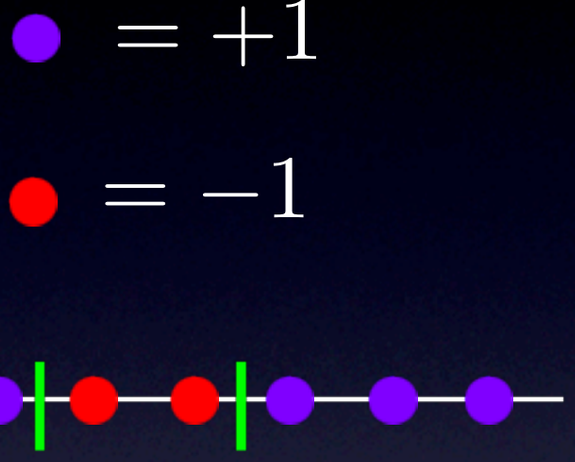
$T > T_C$

rate ~ $\exp(-c\beta J)$





Self-Correcting

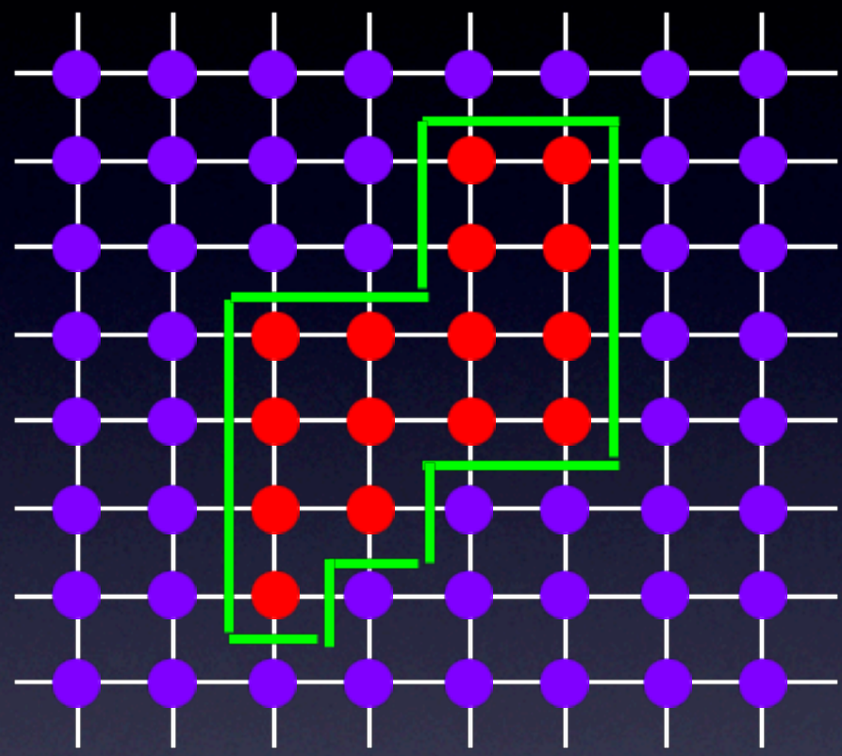


1D



$E \propto$ Constant

2D



$E \propto$ Perimeter

Free energy change to flip all spins

$\Delta F \approx 4J - TN$

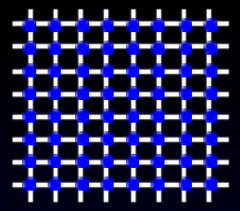
entropy wins

$\Delta F \approx 8N^{\frac{1}{2}}J - T\ln(N \times 2^{2N})$

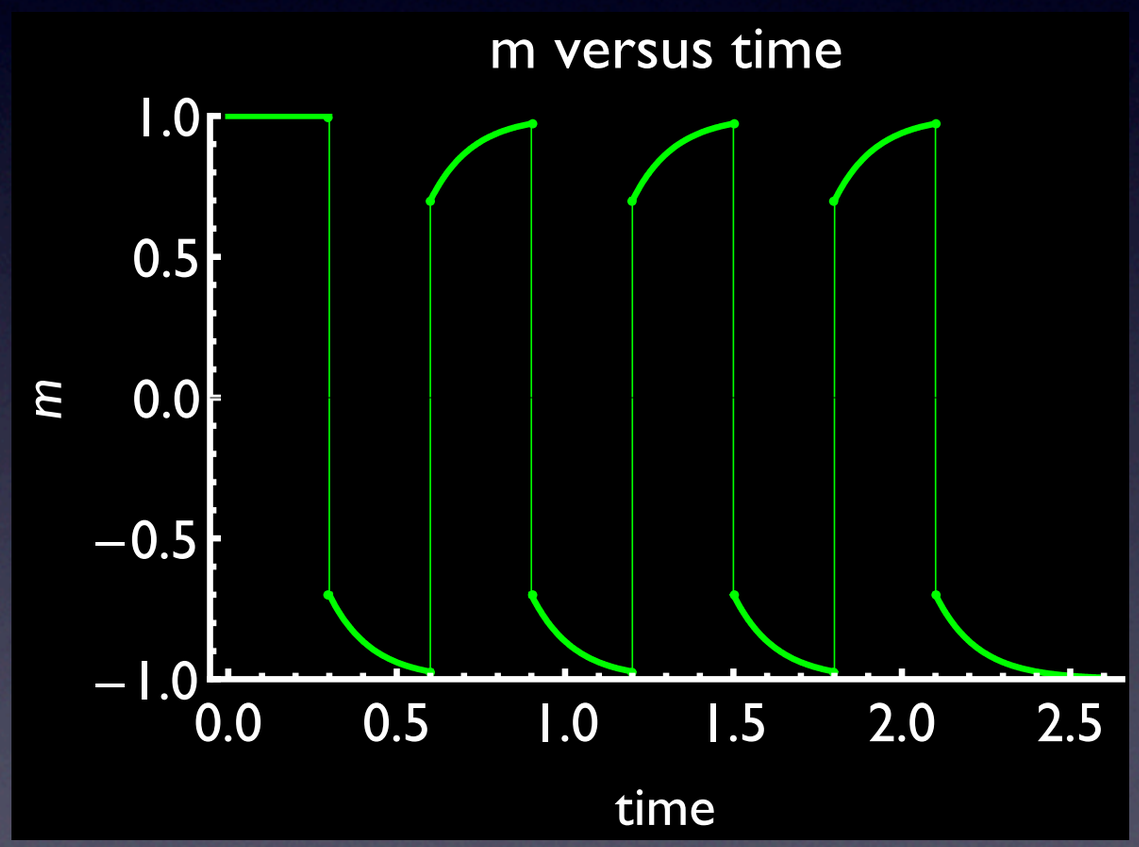
$T < T_C$ energy wins

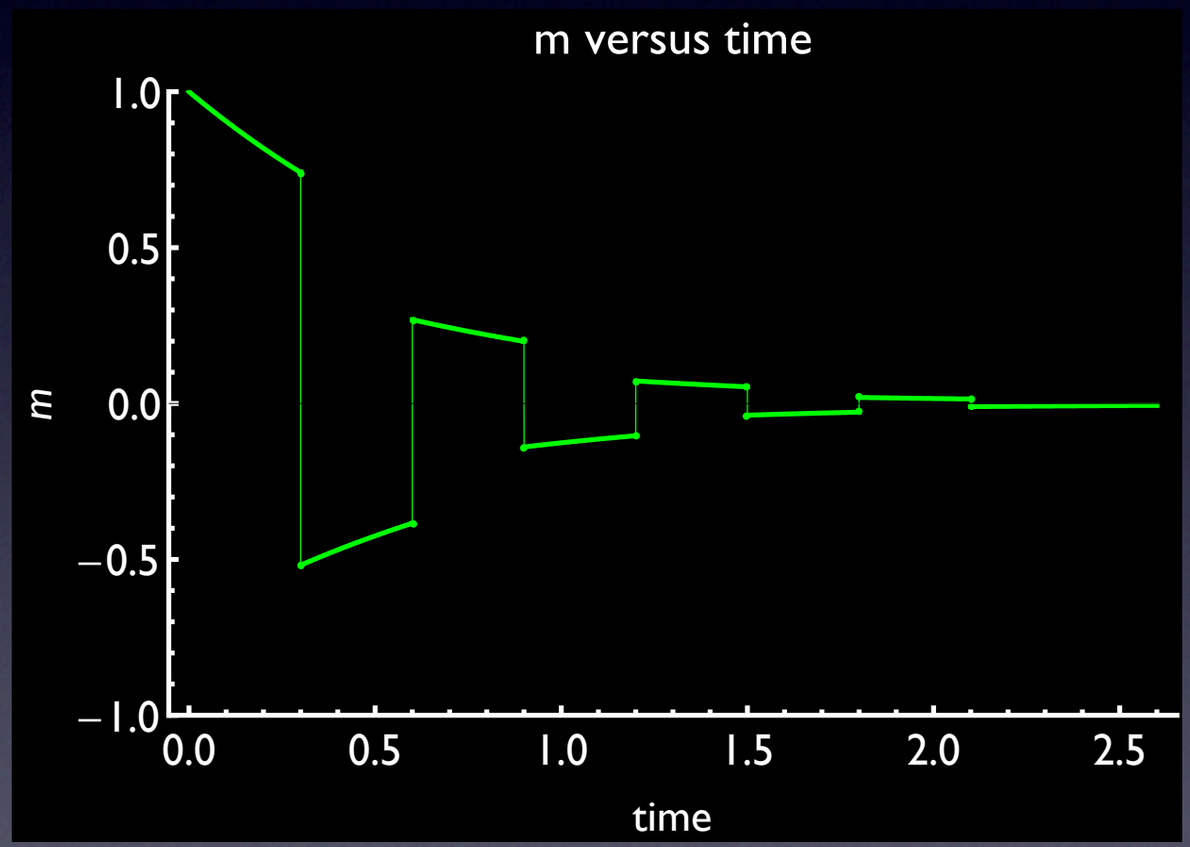
$T > T_C$ entropy wins

Fault-Tolerance

2D 

1D



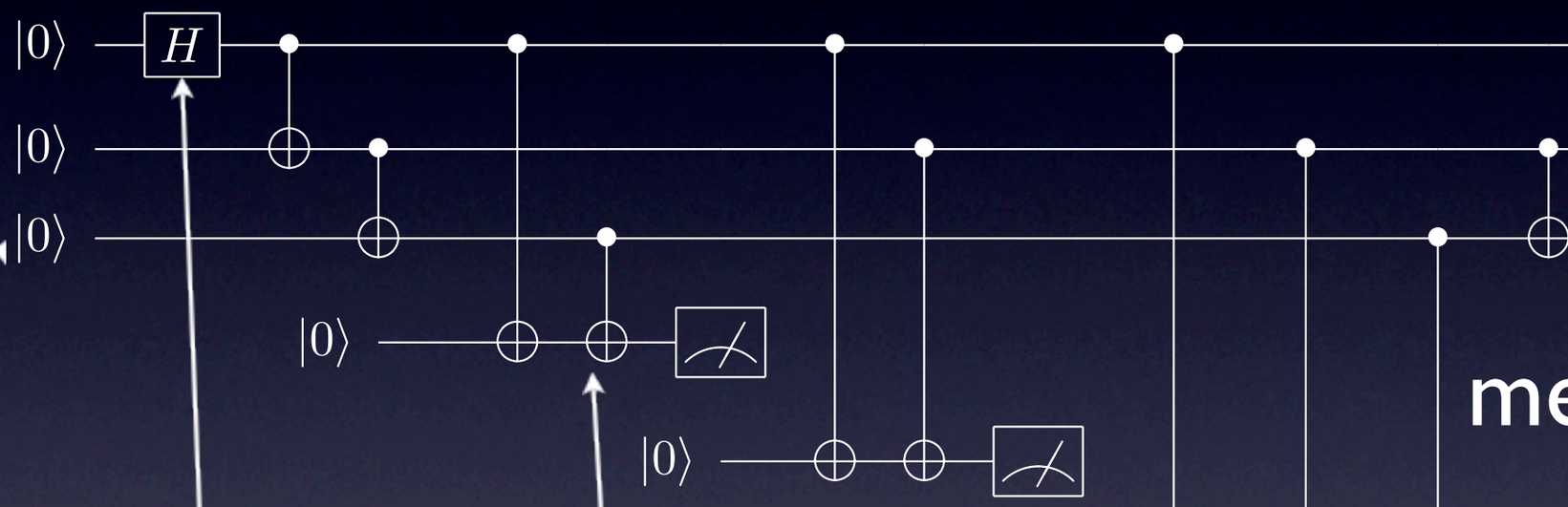




Dave?
Weren't you going to be talking about quantum computing?

Quantum Computing

Circuit model for quantum algorithms



state preparation

quantum gate

quantum gate

measurement

$$H = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$

$$C_X = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

CS time ⟶

Quantum Algorithms

Factor

1881988129206079638386972394616504398071635633794173827007633564229888597152346654853190606065047430453173880113033967161969232120573403187955065699622130516875930765025705

=

3980750864240649373971255005503864911990643623425267084063851895759463889572617685833317

x

4727721461074353025362307197304822463291469530209711645985217113052071125636359039752 7

	Running time
Best classical algorithm	$\exp(N^{1/3})$
Shor's 1994 quantum algorithm	N^3

Breaks public key cryptosystems (RSA, Elliptic curves)

Obstacles

Decoherence:

$$(\alpha|0\rangle + \beta|1\rangle) \otimes |\psi_{\text{environment}}\rangle \rightarrow \alpha^*|0\rangle \otimes |\psi_0\rangle + \beta^*|1\rangle \otimes |\psi_1\rangle$$



pure



mixed

Infinite precision needed for gates?

$$H = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$



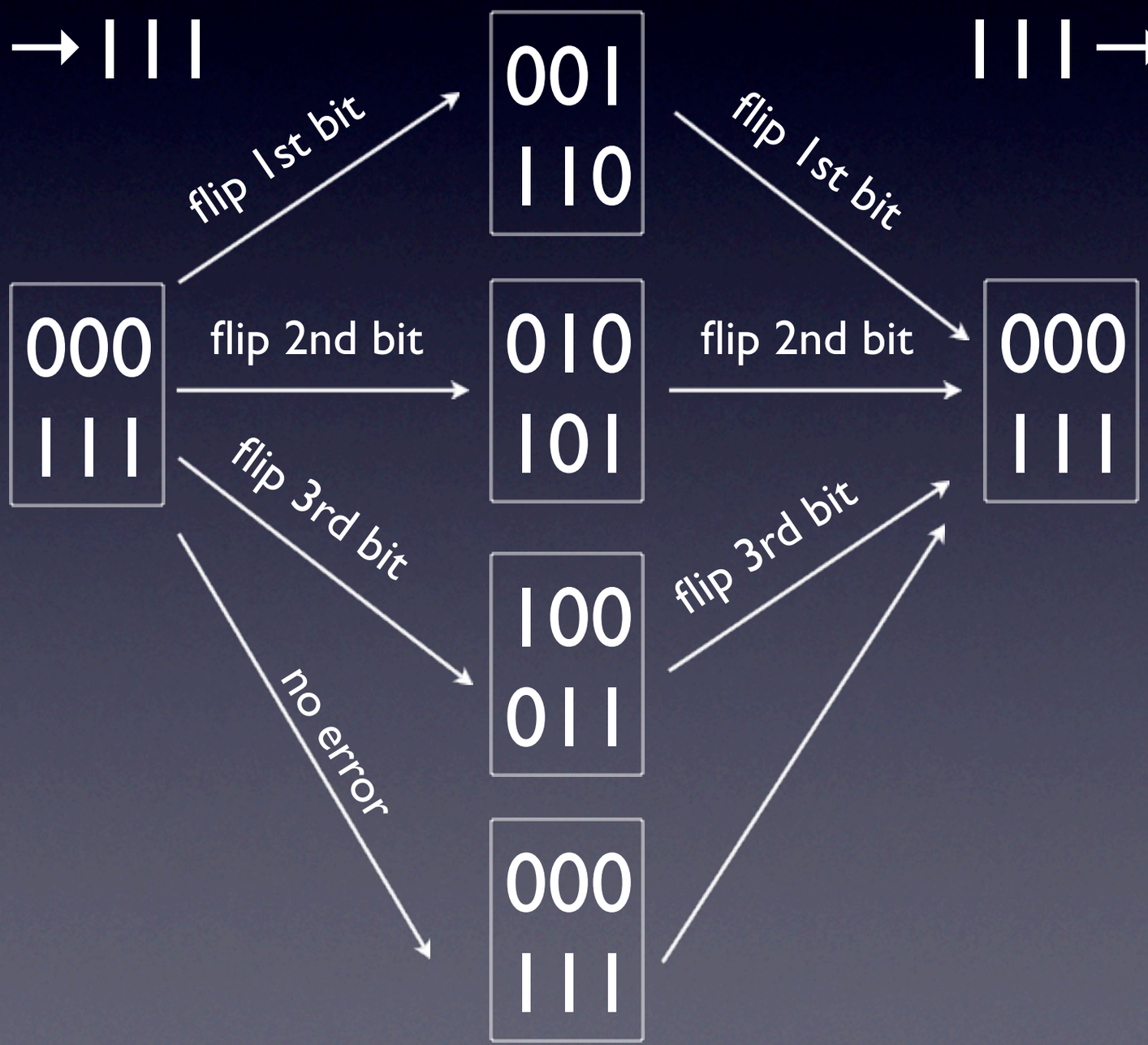
my laser
doesn't have a setting for one
over square root of two

Error Correction

encode
0→000
1→111

diagnose

decode
000→0
111→1



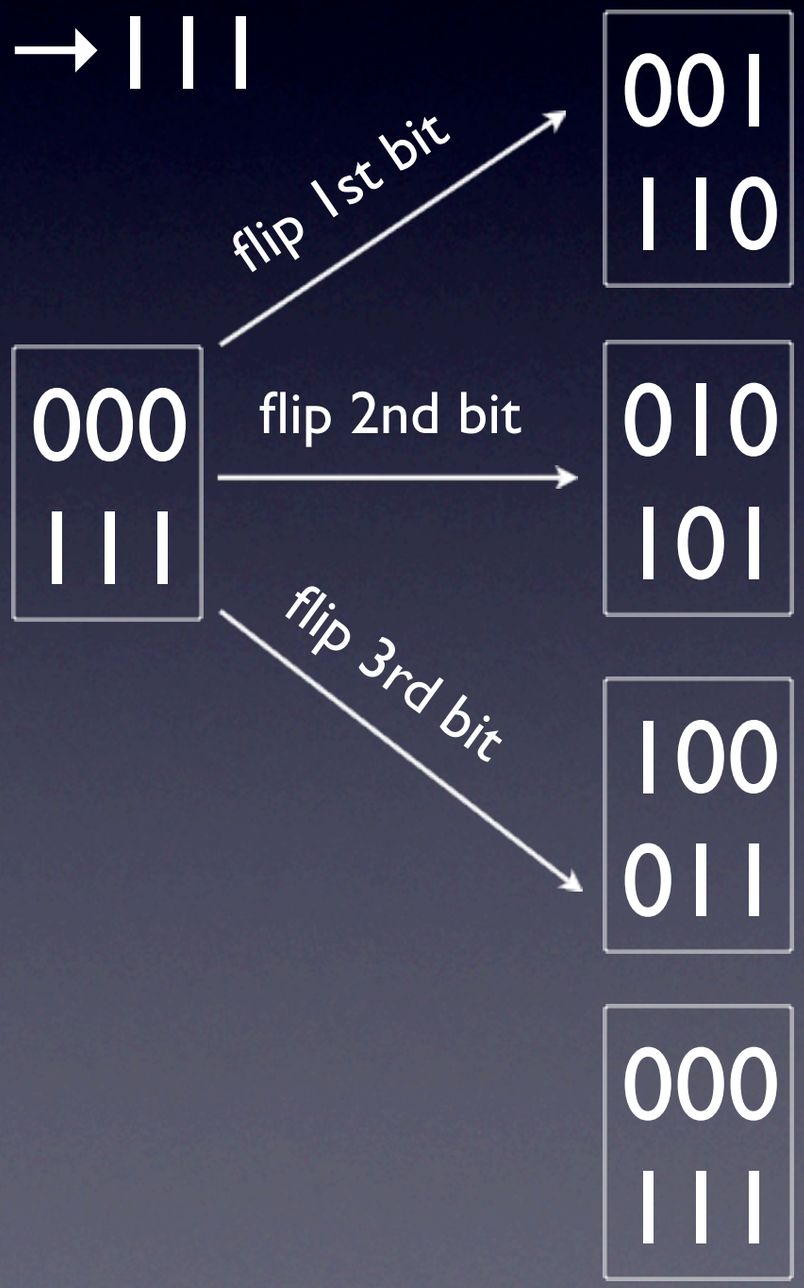
Each bit flipped indepdently with probability p

$p_{fail}=3p^2-p^3$

If $p<1/2$, then $p_{fail}<p$...success!

Error Correction

encode
0→000
1→111



	parity 1 and 2	parity 2 and 3
001 110	0	1
010 101	1	1
100 011	1	0
000 111	0	0

diagnose which error occurred without revealing value of encoded information

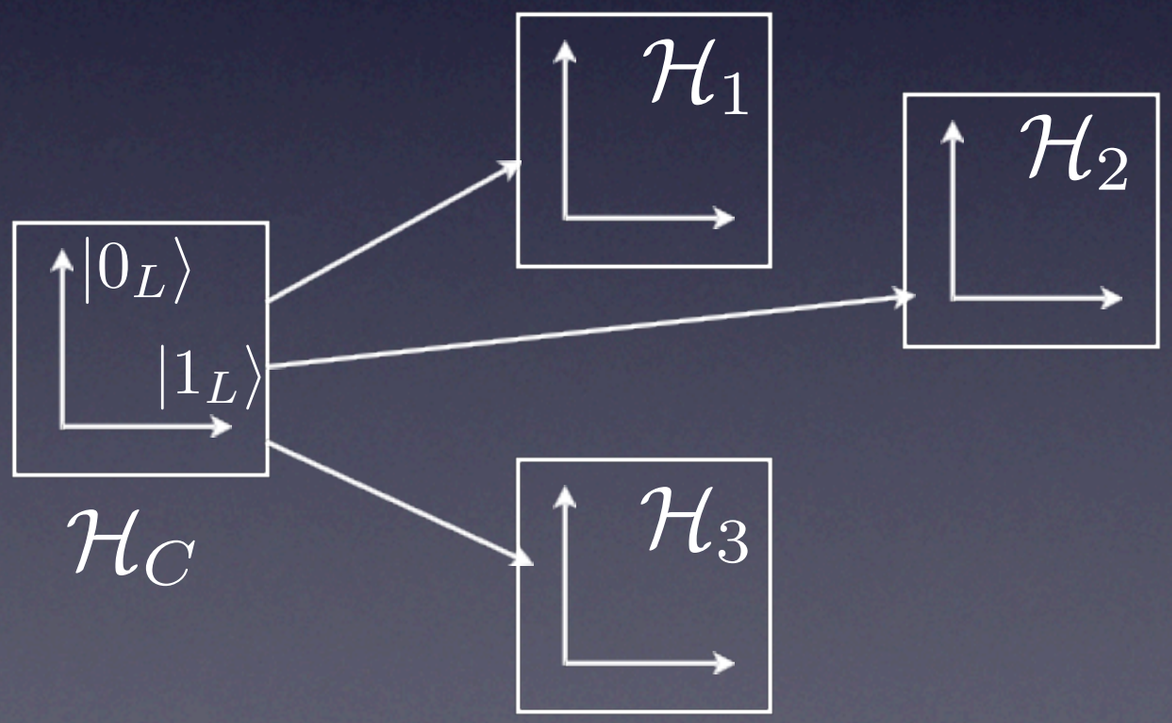
Quantum Error Correction

Basic idea: encoding information can help protect it



$$\mathcal{H} = [\mathbb{C}^2] \otimes [\mathbb{C}^2] \otimes [\mathbb{C}^2] \otimes [\mathbb{C}^2] \otimes [\mathbb{C}^2]$$

error correcting subspace $\mathcal{H}_C \subset \mathcal{H}$



errors take codespace to orthogonal subspaces

$$\mathcal{H}_C \perp \mathcal{H}_1 \perp \mathcal{H}_2 \perp \mathcal{H}_3$$

Quantum Error Correction



$$\mathcal{H} = [\mathbb{C}^2] \otimes [\mathbb{C}^2] \otimes [\mathbb{C}^2] \otimes [\mathbb{C}^2] \otimes [\mathbb{C}^2]$$

Commuting set of four operators

$$S_1 = X \otimes Z \otimes Z \otimes X \otimes I \qquad [S_i, S_j] = 0$$

$$S_2 = I \otimes X \otimes Z \otimes Z \otimes X \qquad S_i^2 = I$$

$$S_3 = X \otimes I \otimes X \otimes Z \otimes Z$$

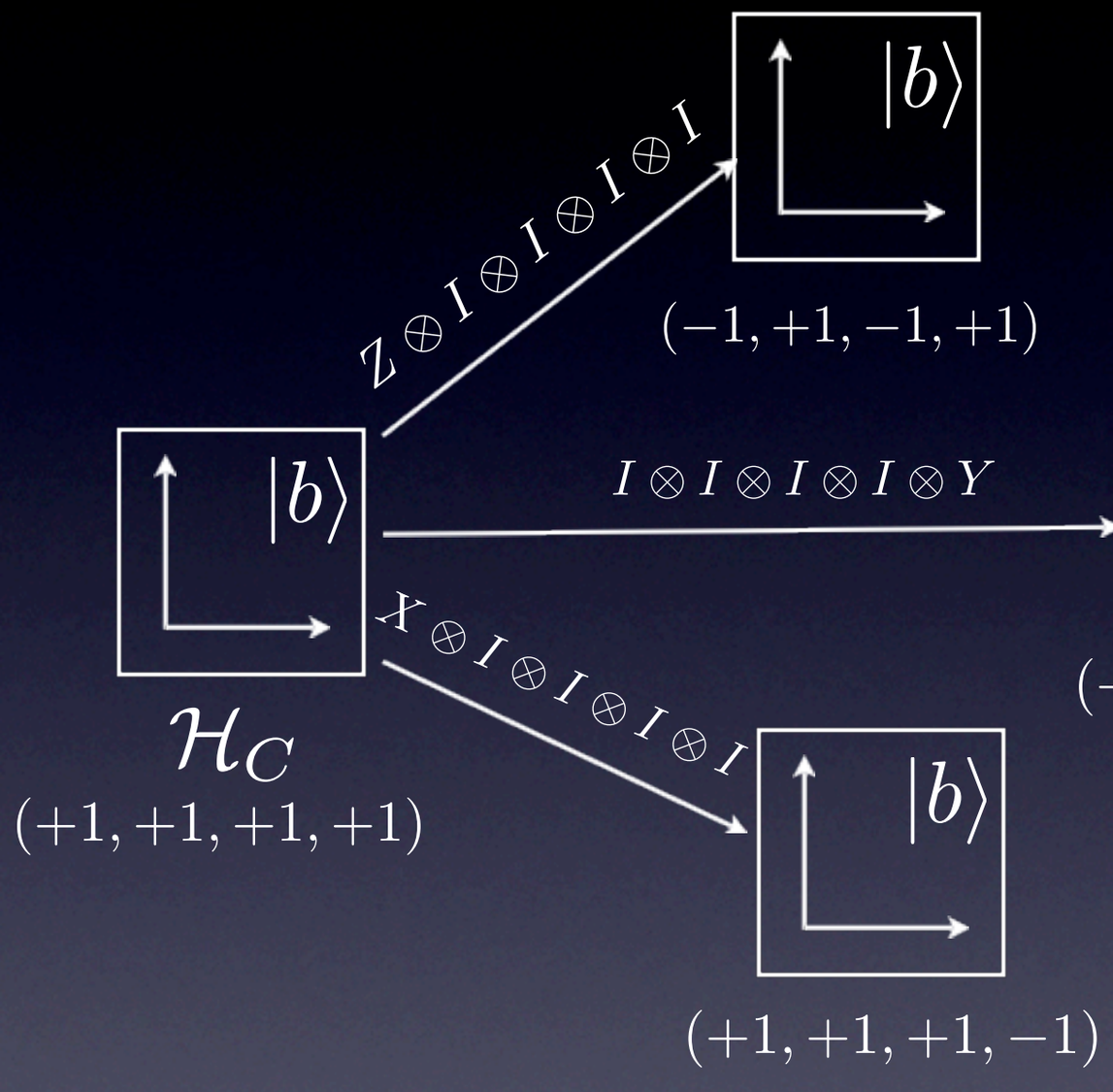
$$S_4 = Z \otimes X \otimes I \otimes X \otimes Z$$

Basis: $|s_1, s_2, s_3, s_4, b\rangle \qquad s_i \in \{+1, -1\}, b \in \{+1, -1\}$

Codespace: $S_i|\psi_C\rangle = |\psi_C\rangle, |\psi_C\rangle \in \mathcal{H}_C$

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Quantum Error Correction

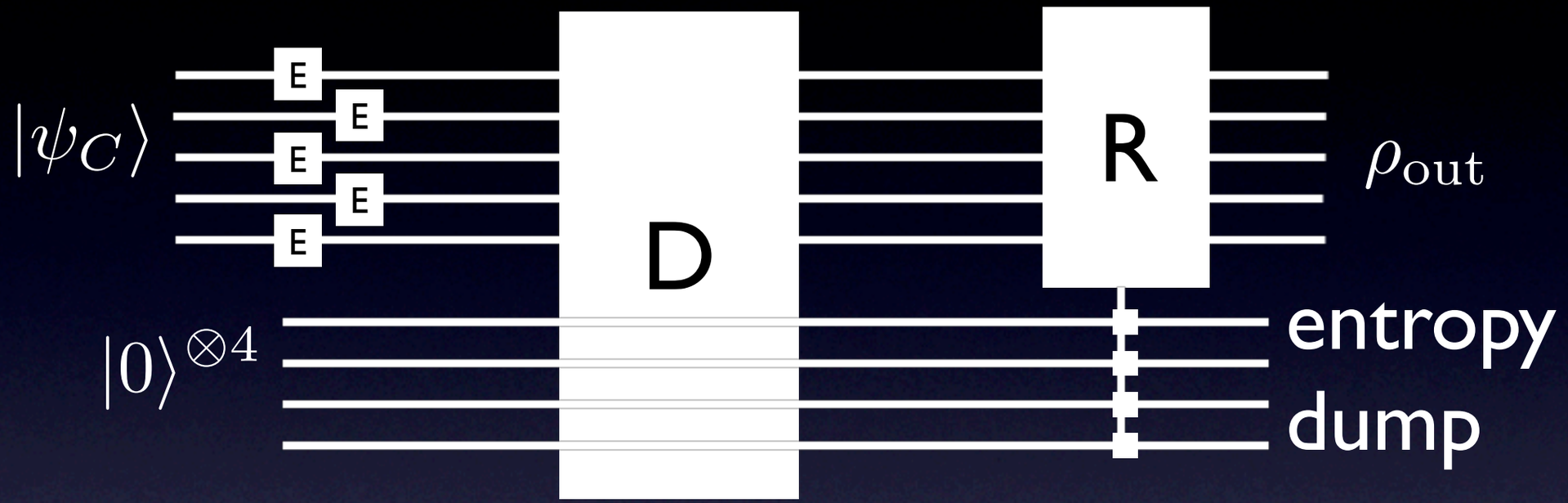


$$S_1 = X \otimes Z \otimes Z \otimes X \otimes I$$
$$S_2 = I \otimes X \otimes Z \otimes Z \otimes X$$
$$S_3 = X \otimes I \otimes X \otimes Z \otimes Z$$
$$S_4 = Z \otimes X \otimes I \otimes X \otimes Z$$
$$(s_1, s_2, s_3, s_4)$$

Every operator which acts nontrivially on only one qubit moves to orthogonal subspace

We can diagnose single qubit errors without destroying quantum information by measuring S_i

Quantum Restorative Organ



If local transition error transition amplitudes are ϵ then output states has error amplitude $C\epsilon^2$

Note: a continuous set of error are corrected by the above procedure (quantum theory is linear)

$$\sum_{i=0}^{15} \alpha_i E_i$$

Fault-Tolerant Quantum Computing

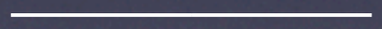
Consider quantum computing where all quantum circuit components can fail:



state preparation



gates

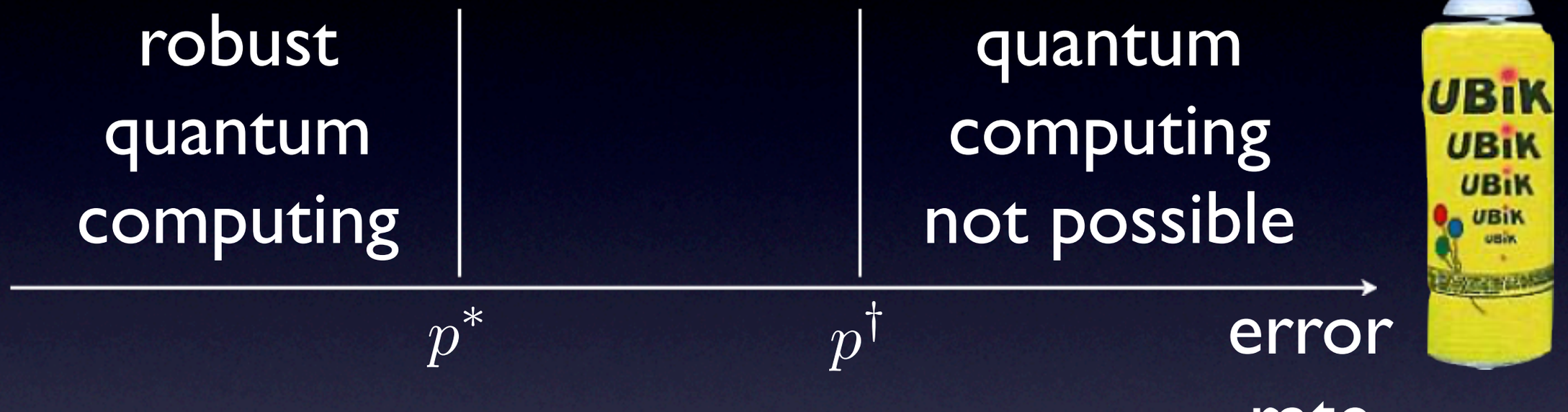


wires



measurements

The Quantum Threshold Theorem



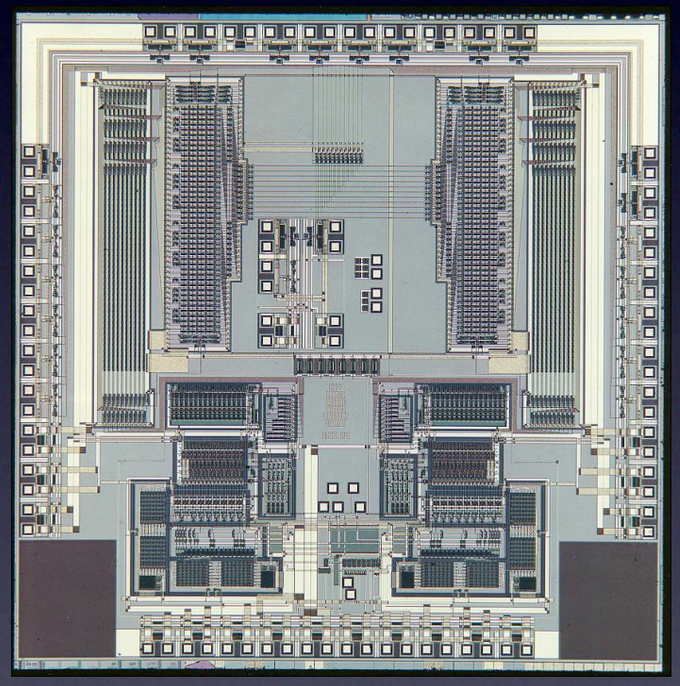


Theorem: If decoherence and lack of control are below a threshold p* (relative to coherent control), then a quantum computation of size g can be carried out with error probability ε using $O(g \log^c(g/\varepsilon))$ extra qubits

Aharonov, Ben-Or, Knill, LaFlamme, Zurek, Shor, Kitaev, Gottesman, Preskill (1996)

But....







Kitaev's Idea

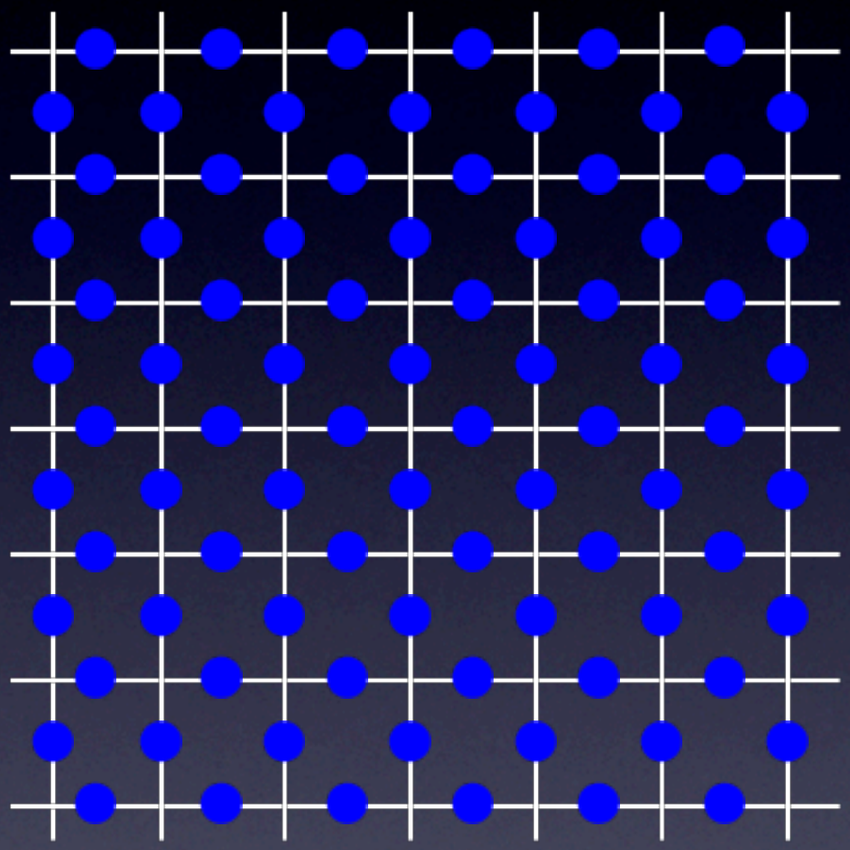


Alexei Kitaev

"Magnetism arise from spins of individual atoms. Each spin is quite sensitive to thermal fluctuations. But the spins interact with each other and tend to be oriented in the same direction. If some spin flips to the opposite direction, the interaction forces it to flip back to the direction of other spins. This process is quite similar to the standard error correction procedure for the repetition code. We may say that errors are being corrected at the physical level. Can we propose something similar in the quantum case? Yes, but it is not so simple."

Kitaev, "Fault-tolerant quantum computation by anyons" (1997, published 2003)

Kitaev's Toric System



periodic boundary conditions (torus)

$\bullet = Z \quad \bullet = X$

plaquette operator



$$S_p = \prod_{e \in \delta P} Z_e$$

vertex operator



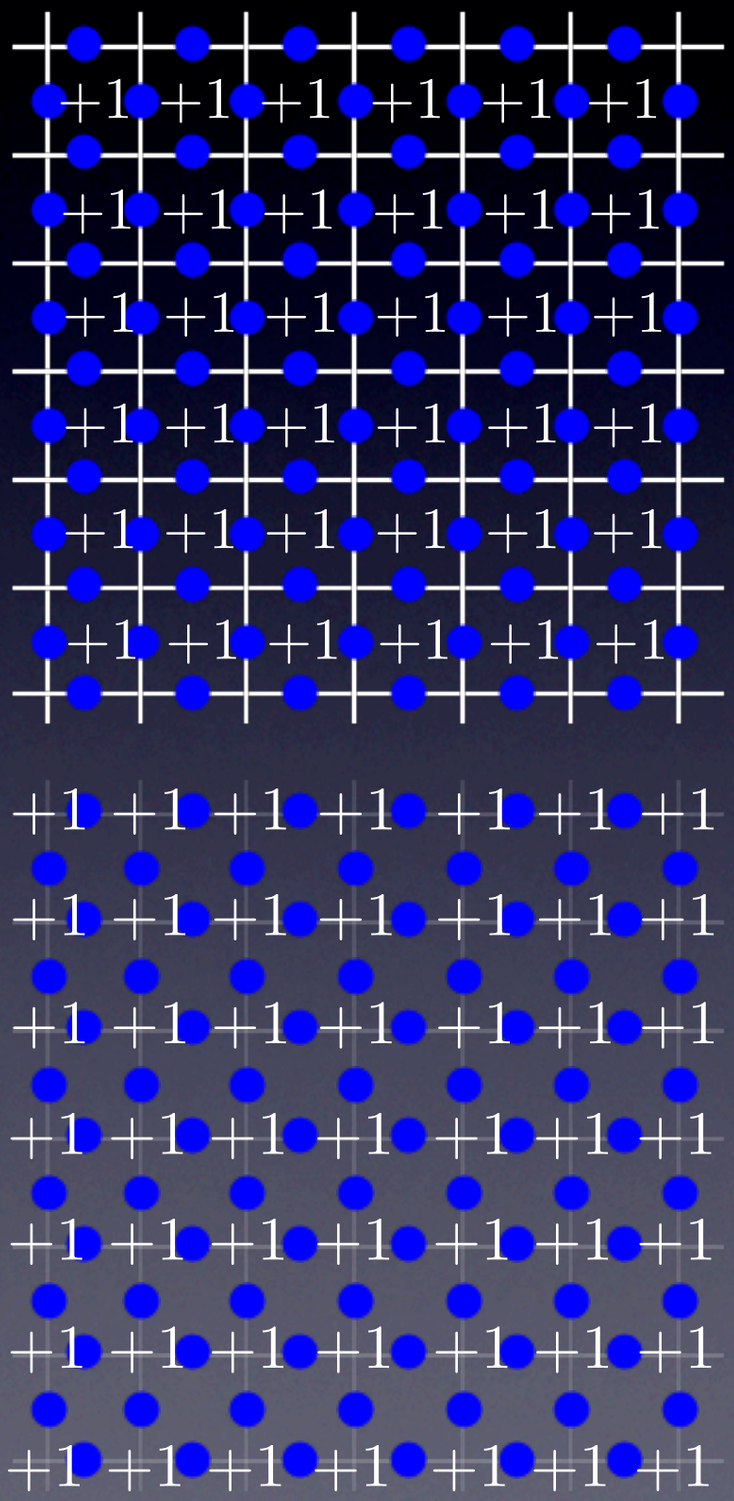
$$S_v = \prod_{e | v \in e} X_e$$

$$[S_p, S_v] = 0 \qquad S_p^2 = S_v^2 = I$$

Kitaev Hamiltonian:

$$H = -\frac{\Delta}{4}\left[\sum_p S_p + \sum_v S_v\right]$$

Kitaev's Toric System



$$H = -\frac{\Delta}{4}\left[\sum_p S_p + \sum_v S_v\right]$$

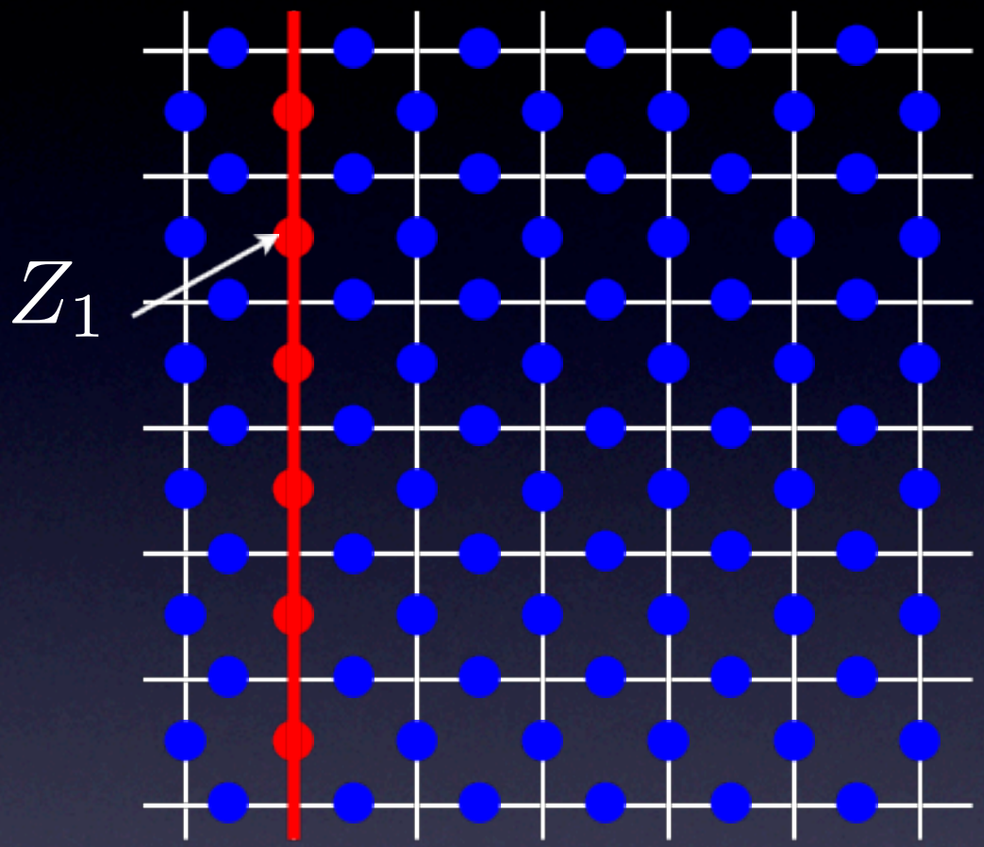
Ground state has

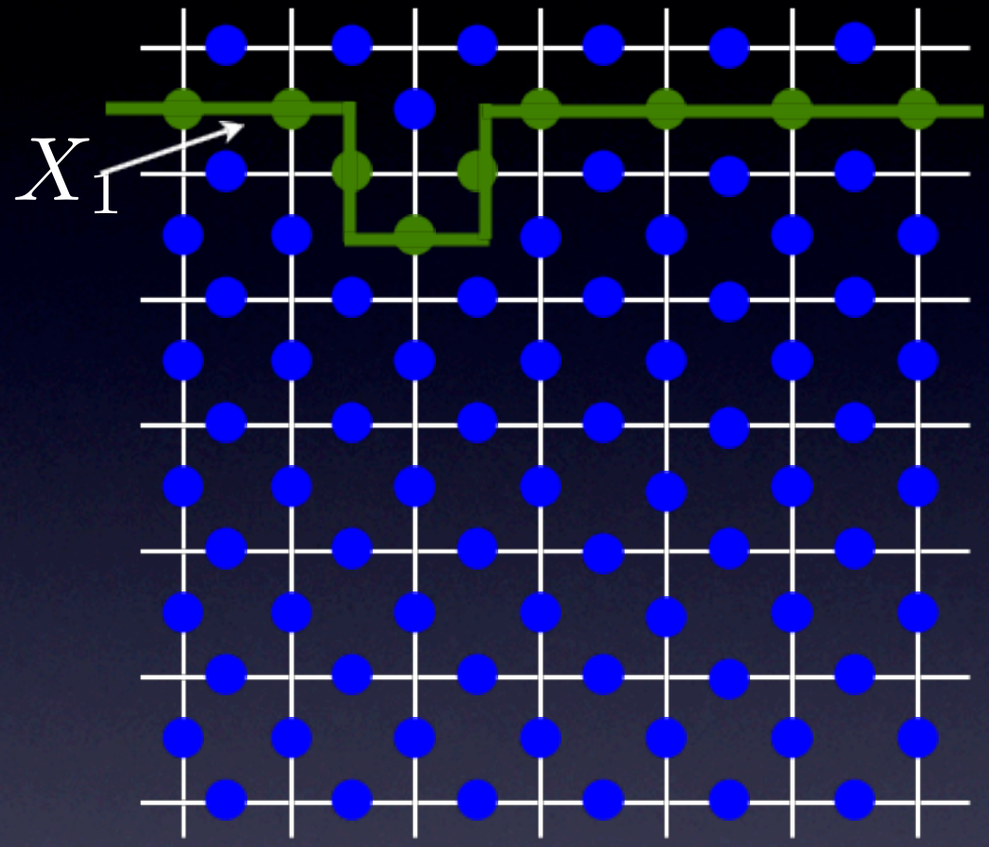
$$S_p|\psi\rangle = |\psi\rangle$$

$$S_v|\psi\rangle = |\psi\rangle$$

Degeneracy?

Kitaev's Toric Qubits



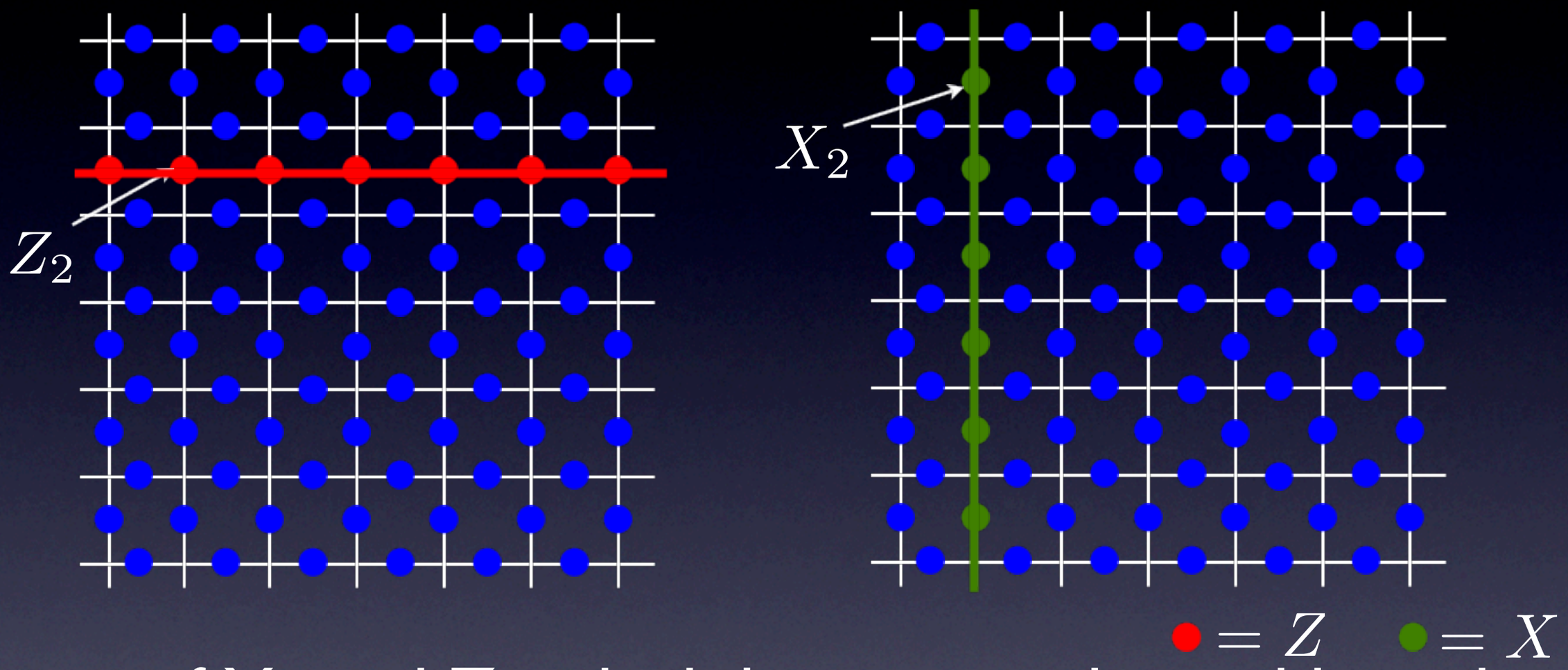


$\bullet = Z \quad \bullet = X$

strings of Xs and Zs which loop around an odd number of times commute with S_p, S_v but cannot be generated as products of these operators

$$[S_p, X_1] = [S_v, X_1] = [S_p, Z_1] = [S_v, Z_1] = 0 \quad [X_1, Z_1] \neq 0$$

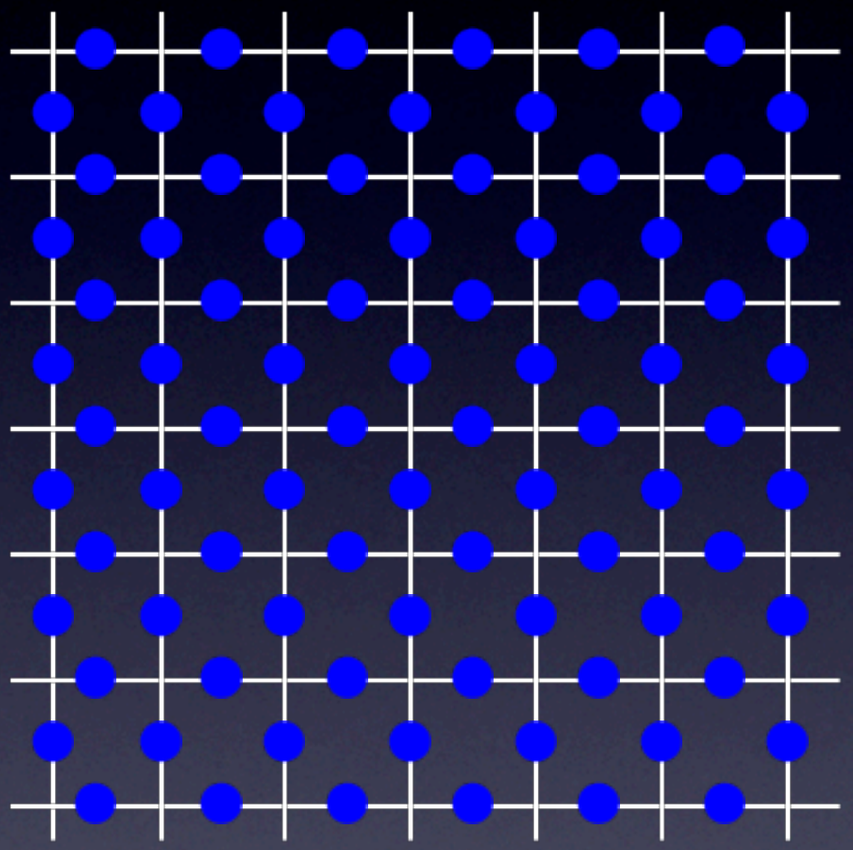
Kitaev's Toric Qubits



strings of Xs and Zs which loop around an odd number of times commute with S_p, S_v but cannot be generated as products of these operators

$$[S_p, X_2] = [S_v, X_2] = [S_p, Z_2] = [S_v, Z_2] = 0 \;\; [X_2, Z_2] \neq 0$$

Kitaev's Toric System



Kitaev Hamiltonian:

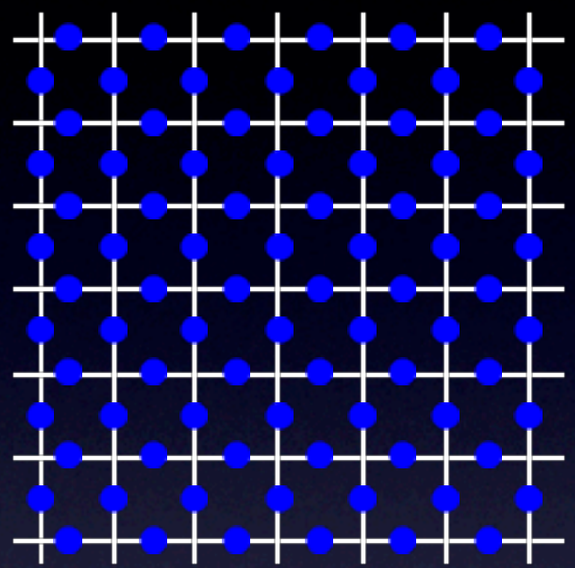
$$H = -\frac{\Delta}{4}\left[\sum_p S_p + \sum_v S_v\right]$$



ground state is a quantum error correcting code

$2L^2$ qubits, encodes 2 qubits, can correct L/2 qubit errors

Kitaev's Toric Protection



Kitaev Hamiltonian:

$$H = -\frac{\Delta}{4}\left[\sum_p S_p + \sum_v S_v\right]$$

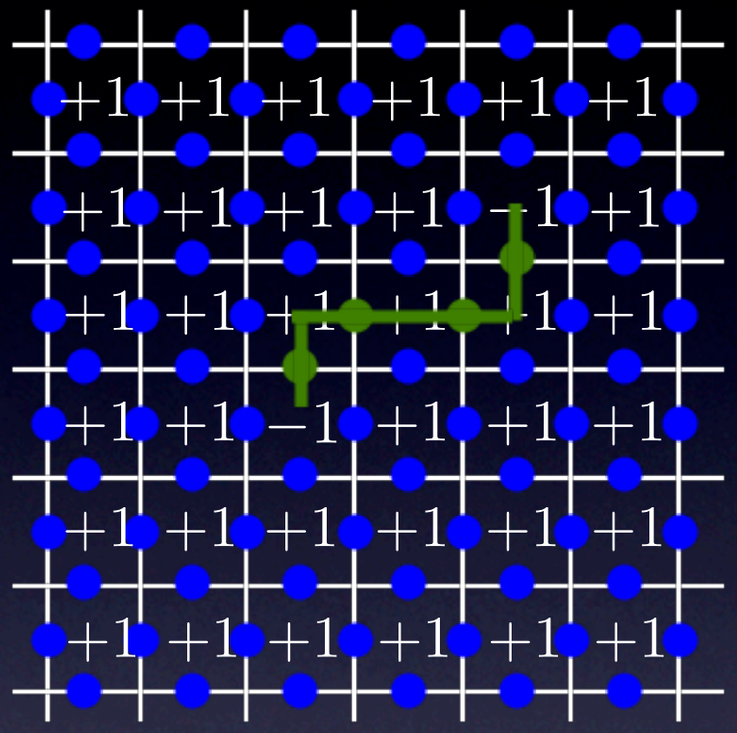
$$\frac{}{2\Delta}$$

Single qubit perturbation $V = \lambda \sum_e P_e$

$$\delta E = \frac{\lambda^L}{\Delta^{L-1}}$$

Exponential suppression of level splitting due to external (local) perturbations

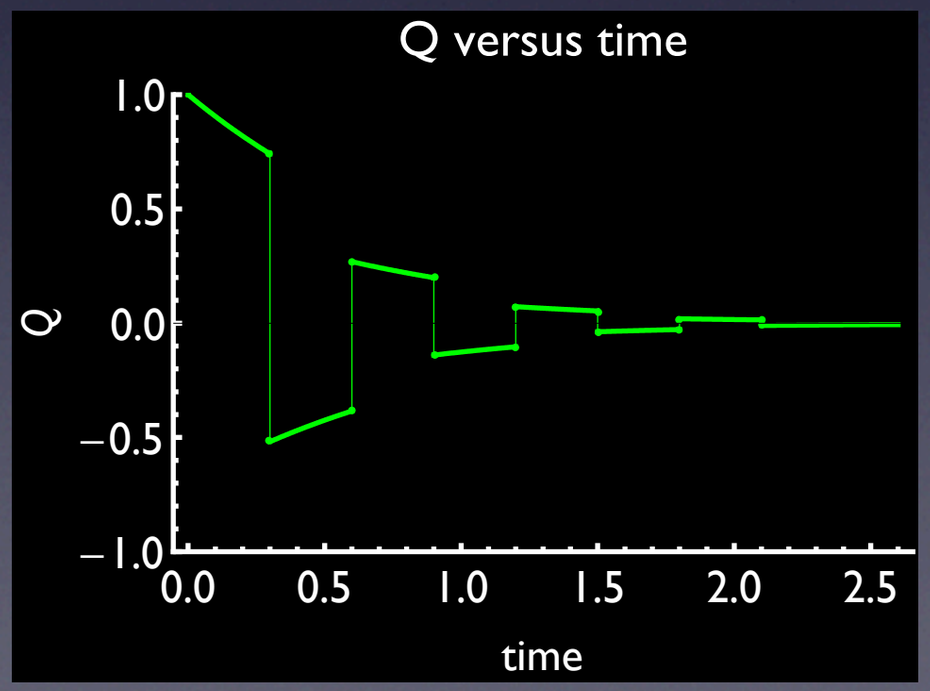
Kitaev's Toric Protection



End of chain of X operators on dual lattice only changes S_p at the end of the chain

Excitations are particles

Like 1D Ising model: good for memory but not self correcting



Topological Quantum Computing



non-abelian Anyons can be used to build a universal quantum computer

$$H = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$



quantum computing using representations of braid group

Kitaev, "Fault-tolerant quantum computation by anyons" (1997, published 2003)
Freedman, "Quantum computation and the localization of modular functors" (2001)

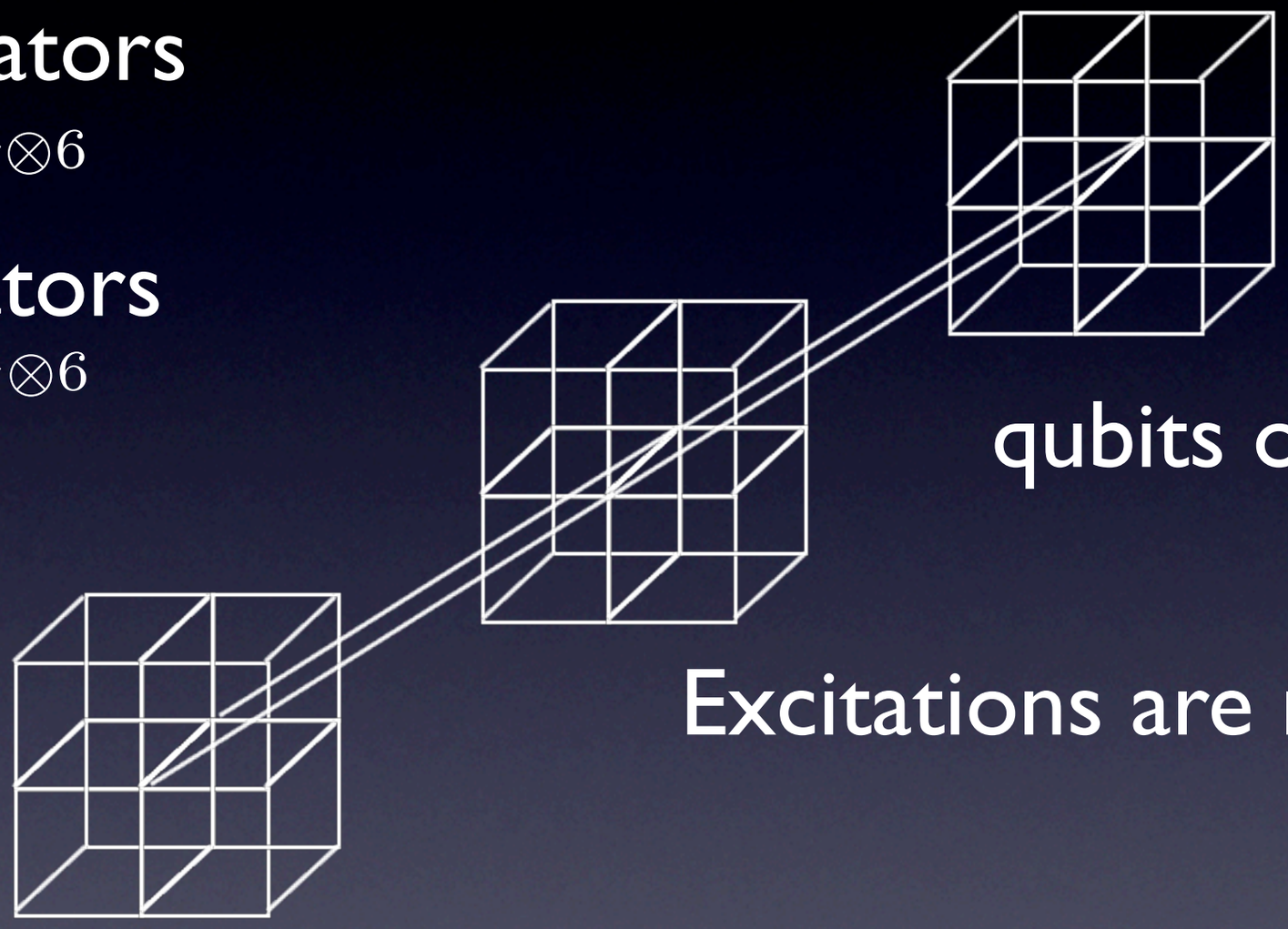
4D Toric Code

cube operators

$$S_c = Z^{\otimes 6}$$

link operators

$$S_l = X^{\otimes 6}$$



qubits on plaquettes

Excitations are now strings

4D code is self-correcting

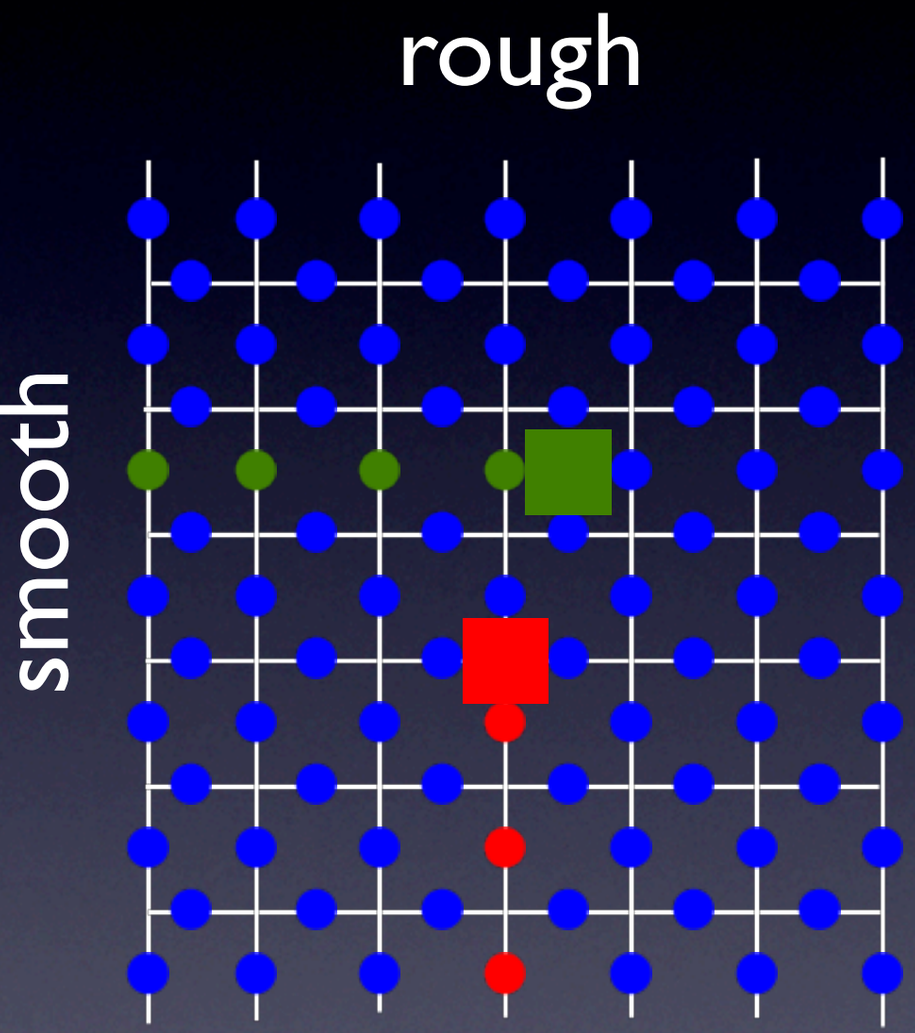
Dennis, Kitaev, Landahl, Preskill “Topological quantum memory” (2002)



Challenge: Does there exist a self-correcting quantum system in 3 spatial dimensions?

and in the meantime....the computer engineers...

Surface Code



Z (X) particles can fall off rough (smooth) boundaries

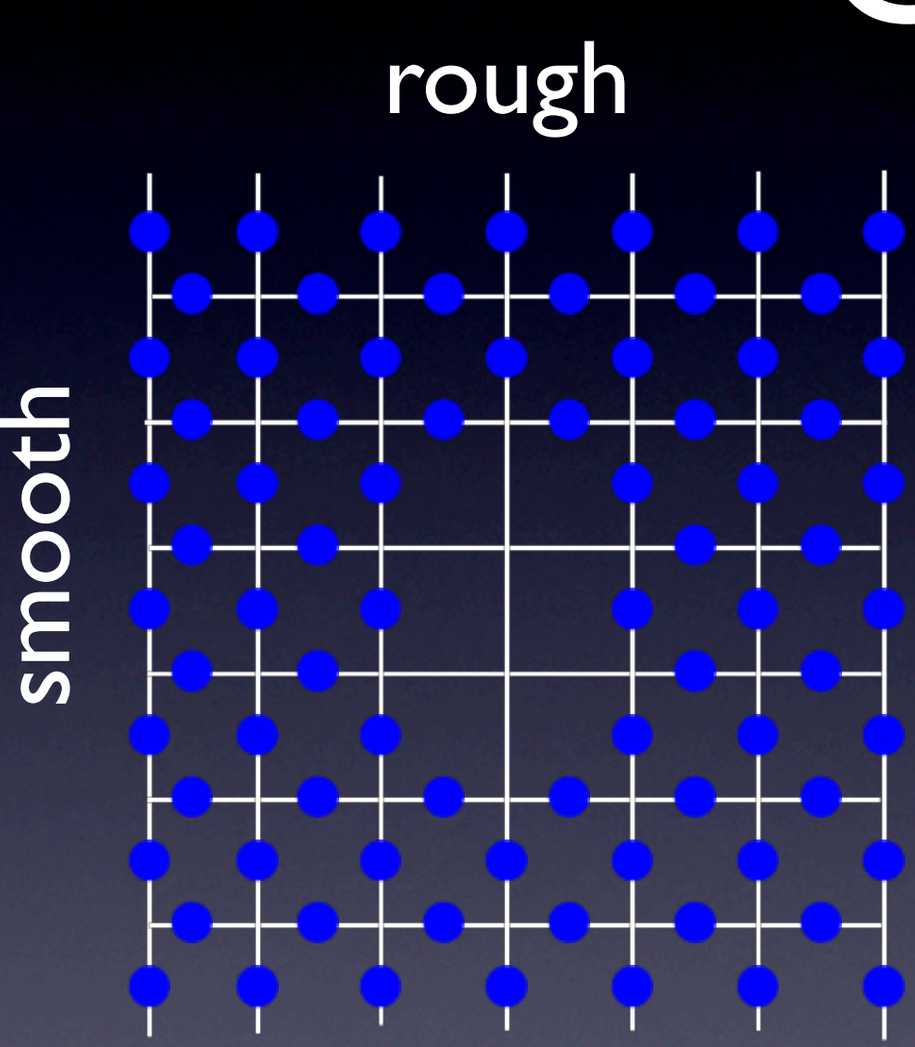
Use as a code not as a physical system

Advantage: no torus! local measurements!

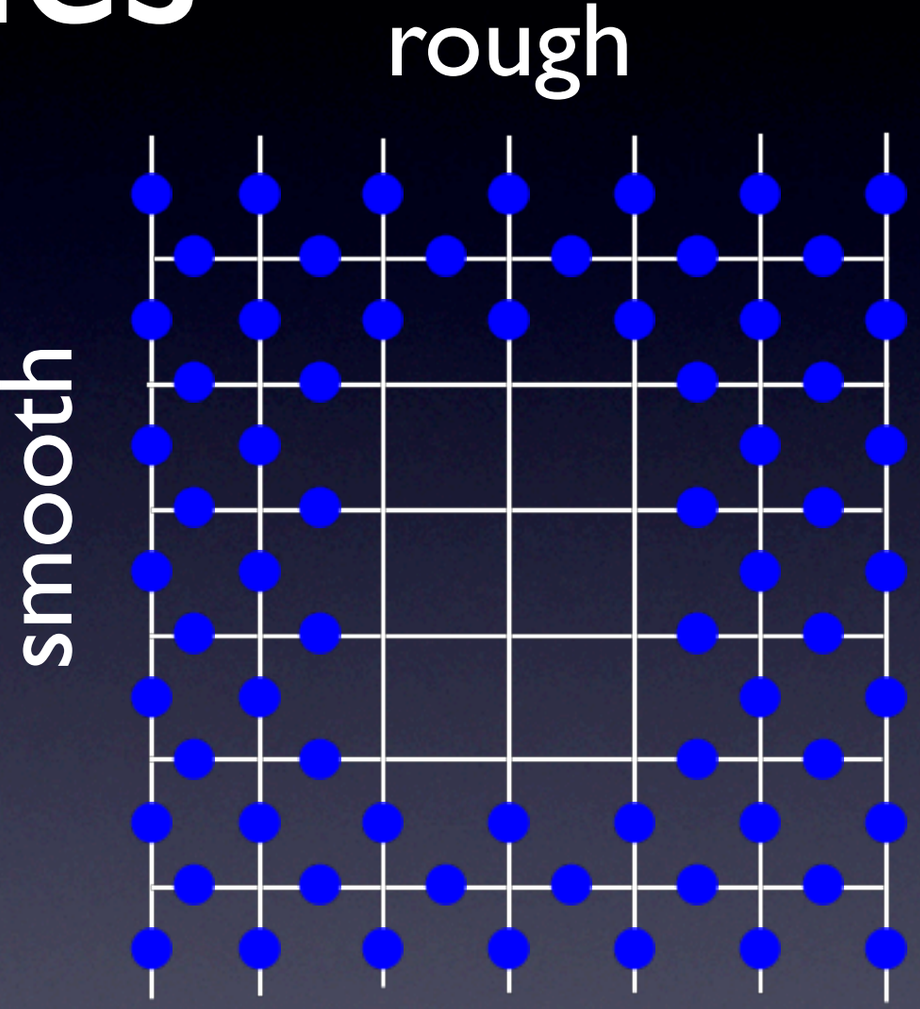
Memory threshold ~ 10^{-4}

Dennis, Kitaev, Landahl, Preskill "Topological quantum memory" (2002)

Puncturing Surface Codes

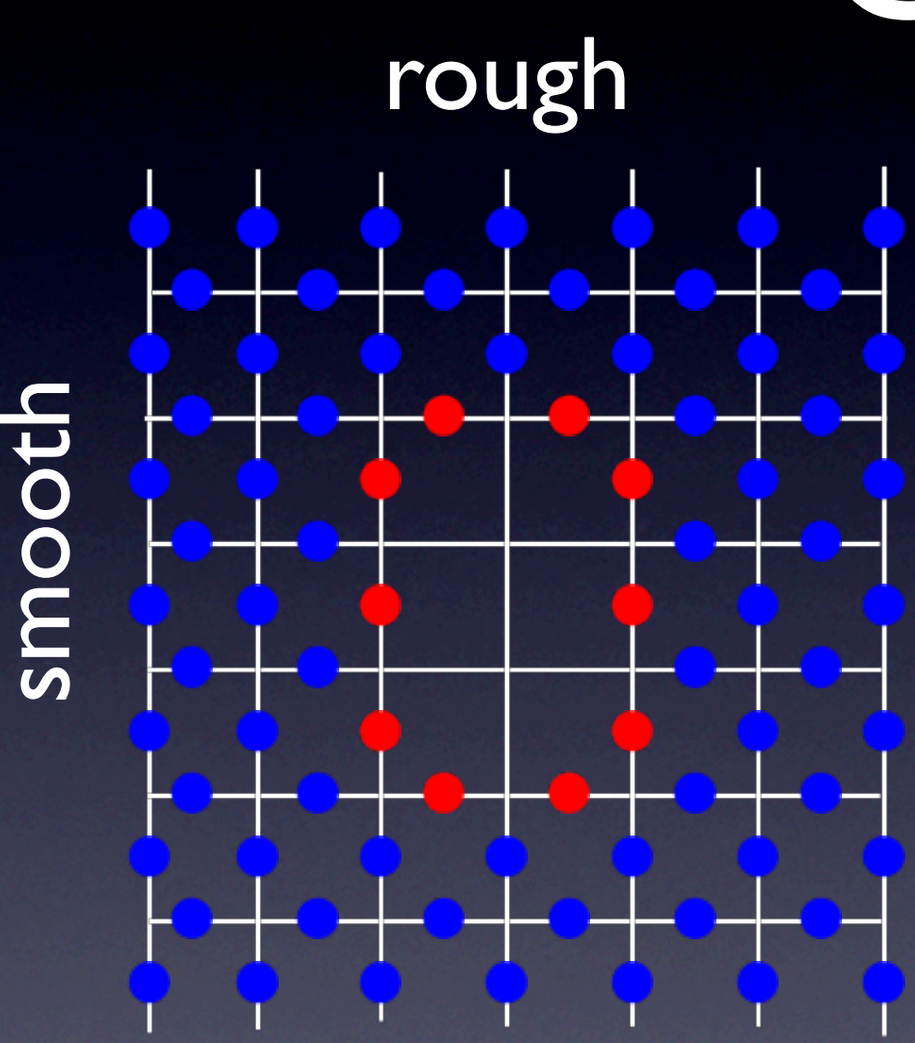


smooth defect

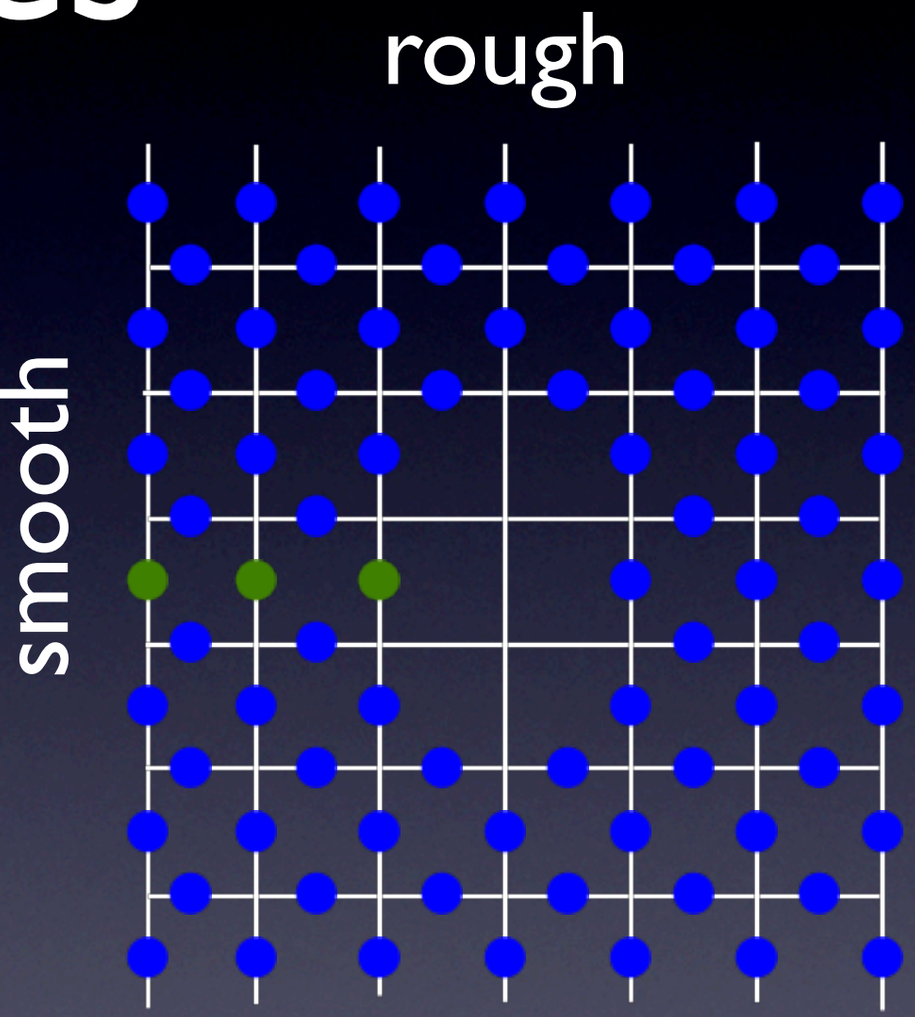


rough defect

Puncturing Surface Codes



Z_L

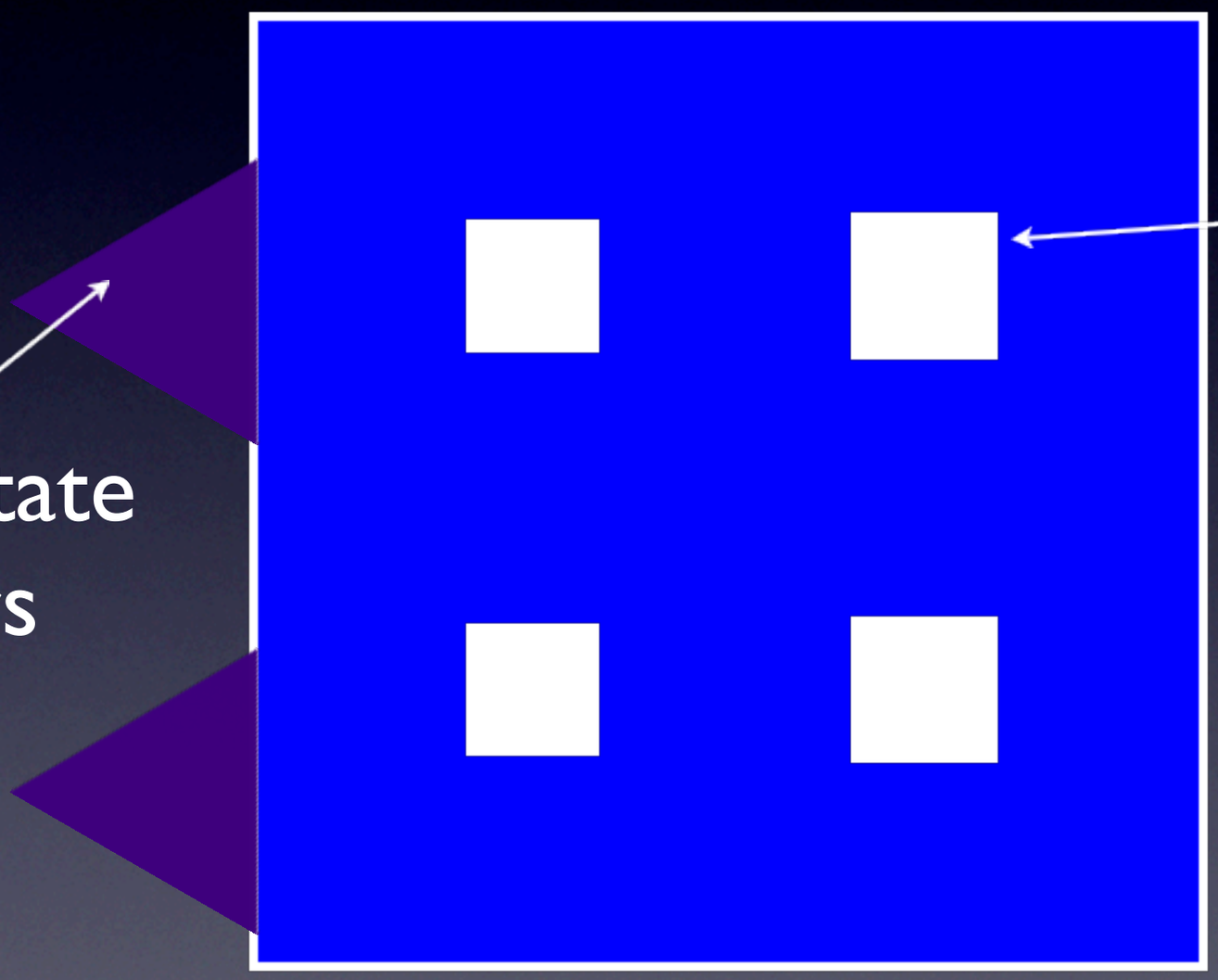


X_L

smooth defect

Local Architecture

Build a nearly local quantum error correcting architecture:



The Computer Engineers have hijacked topological quantum computing!

Subsystem Codes

Take a subspace $\mathcal{H}_S \subset \mathcal{H}$ and put a tensor product on it

$$\mathcal{H}_S = \mathcal{H}_L \otimes \mathcal{H}_G$$

encode quantum information into $\mathcal{H}_L$

One doesn't care what happens to information in $\mathcal{H}_G$

Quantum information protected up to change in "gauge"

Stabilizer Codes

Pauli group on n qubits: $\langle i^k P_1 \otimes P_2 \otimes \cdots P_n \rangle$

$P_i \in \{I, X, Y, Z\}$

Stabilizer group: Abelian subgroup of Pauli group not containing $-I$

$$\mathcal{S} = \langle S_1, S_2, \ldots, S_{n-k} \rangle$$

Stabilizer code: subspace fixed by stabilizer group

$$S_i |\psi_C\rangle = |\psi_C\rangle$$

Normalizer of stabilizer in Pauli group: logical operators

$$\mathcal{N} \quad \begin{matrix} X_1 & X_2 & \cdots & X_k \\ Y_1 & Y_2 & \cdots & Y_k \\ Z_1 & Z_2 & \cdots & Z_k \end{matrix}$$

[[n,k,d]] code

Stabilizer Subsystem Codes

Normalizer of stabilizer in Pauli group:

$$\mathcal{N} \quad \begin{matrix} X_1 & X_2 & \cdots & X_k \\ Y_1 & Y_2 & \cdots & Y_k \\ Z_1 & Z_2 & \cdots & Z_k \end{matrix}$$

Divide into gauge and logical qubits:

$$\mathcal{N} = \mathcal{N}_G \otimes \mathcal{N}_L$$

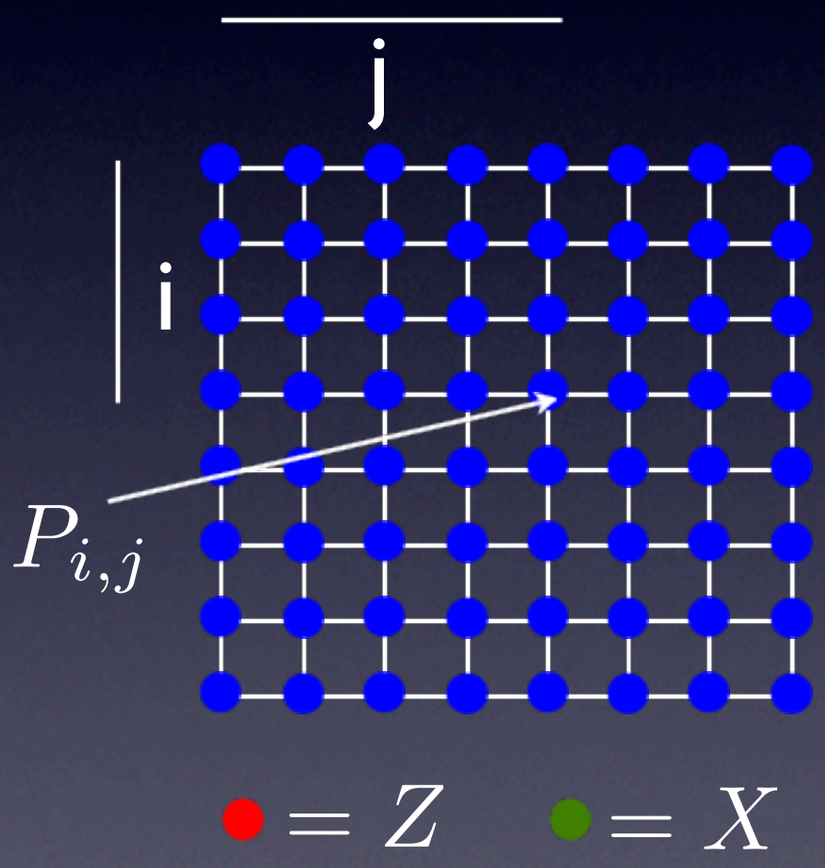
$$\begin{matrix} X_1 & X_2 & \cdots & X_g \\ Y_1 & Y_2 & \cdots & Y_g \\ Z_1 & Z_2 & \cdots & Z_g \end{matrix} \qquad \begin{matrix} X_1 & X_2 & \cdots & X_{k-g} \\ Y_1 & Y_2 & \cdots & Y_{k-g} \\ Z_1 & Z_2 & \cdots & Z_{k-g} \end{matrix}$$

gauge logical

[[n,k,g,d]] subsystem code

Doubling the Toric Code

L x L (L even) square lattice



z type

$$S^z_{i,j} = Z_{i,j} Z_{i+1,j} Z_{i,j+1} Z_{i+1,j+1}$$

x type

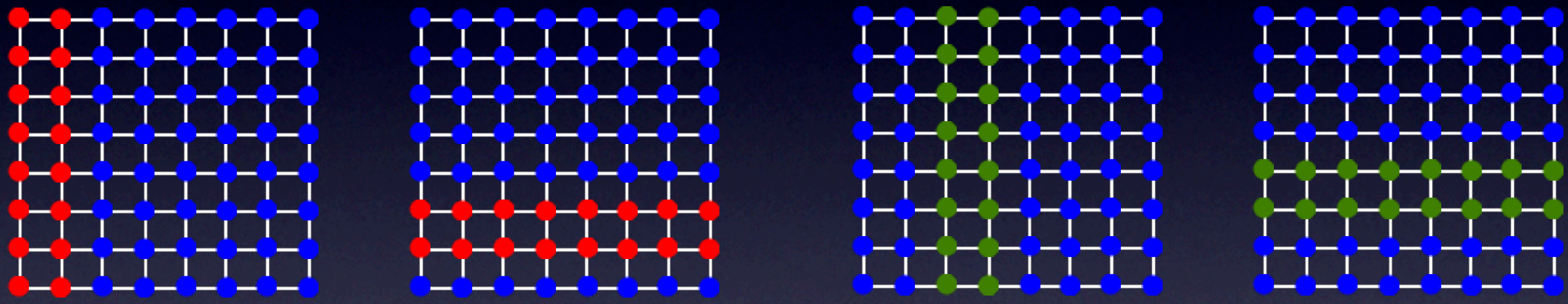
$$S^x_{i,j} = X_{i,j} X_{i+1,j} X_{i,j+1} X_{i+1,j+1}$$

Consider the group generated by these operators:

$$\mathcal{T} = \langle \quad , \quad \rangle$$

Doubled Toric Code

Abelian invariant subgroup:



$$C_j^X = \prod_{i=1}^{L} X_{i,j} X_{i+1,j} \quad C_i^X = \prod_{i=1}^{L} X_{i,j} X_{i,j+1} \quad C_j^Z = \prod_{i=1}^{L} Z_{i,j} Z_{i+1,j} \quad C_i^Z = \prod_{i=1}^{L} Z_{i,j} Z_{i,j+1}$$

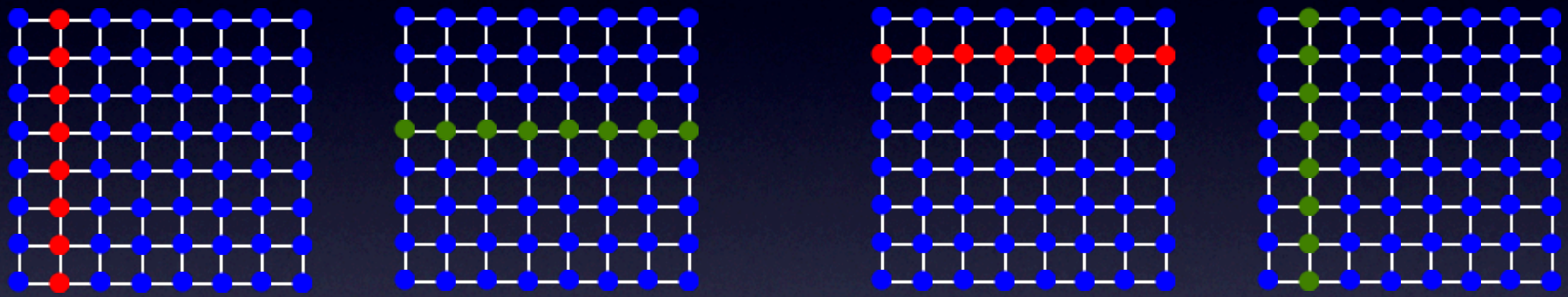
$(L-1)^2 - 1$ generators $(L-1)^2 - 1$ generators

Stabilizer group

● $= Z$ ● $= X$

Doubled Toric Code

Normalizer of stabilizer group: $\mathcal{T}$ and two logical qubits

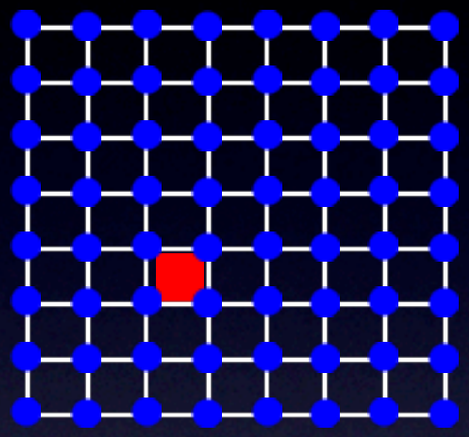


Z_1 X_1 Z_2 X_2

[[L^2,2,L^2-2,L/2]] subsystem code

● $= Z$ ● $= X$

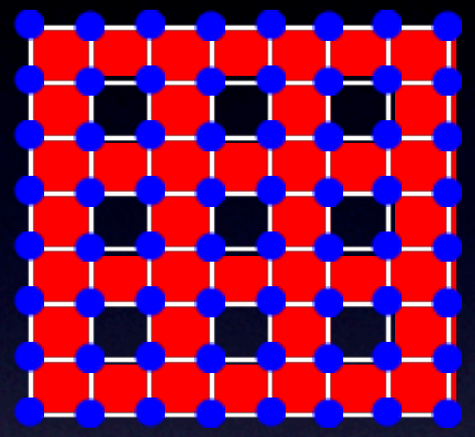
Puncturing the DTC



remove one

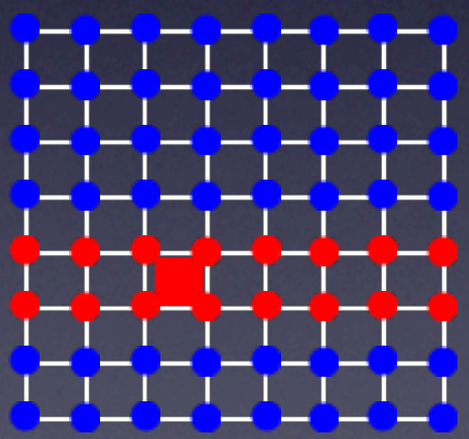


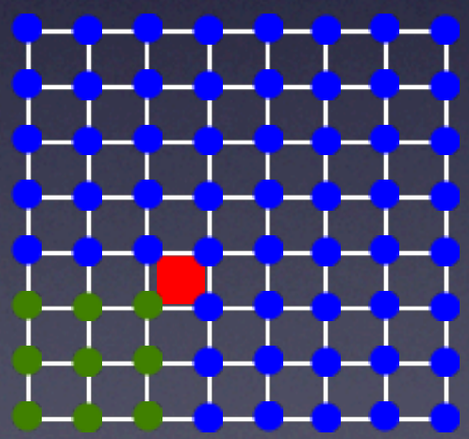
z type



region of allowed removal

Z_L

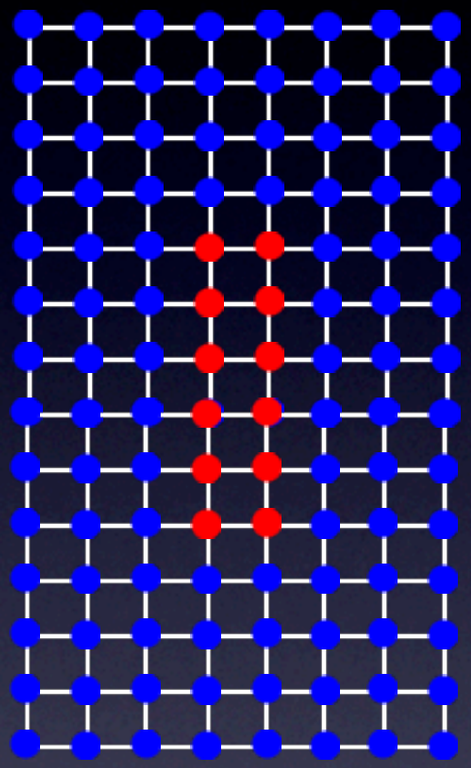




X_L

produces one encoded qubit

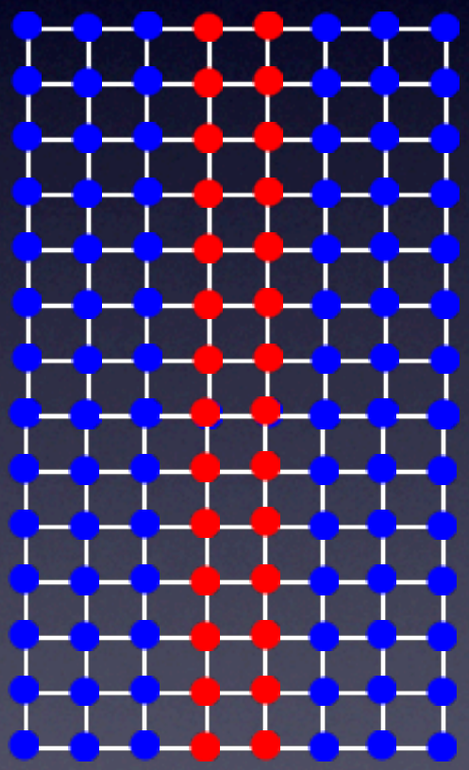
Puncturing the DTC



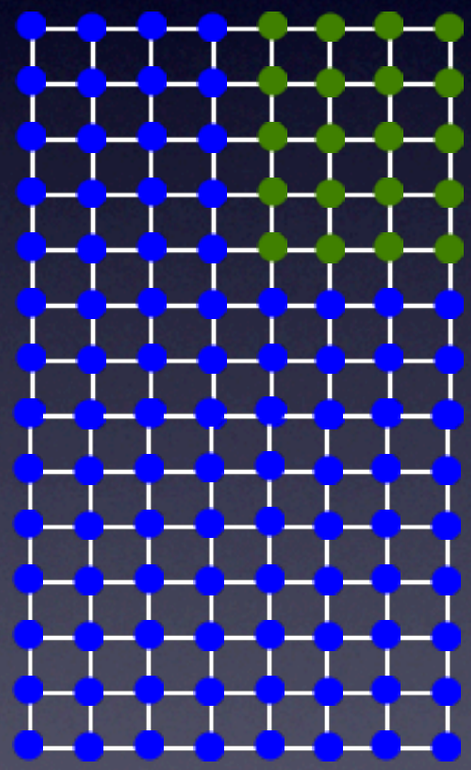
remove > 1?



z type



Z_L

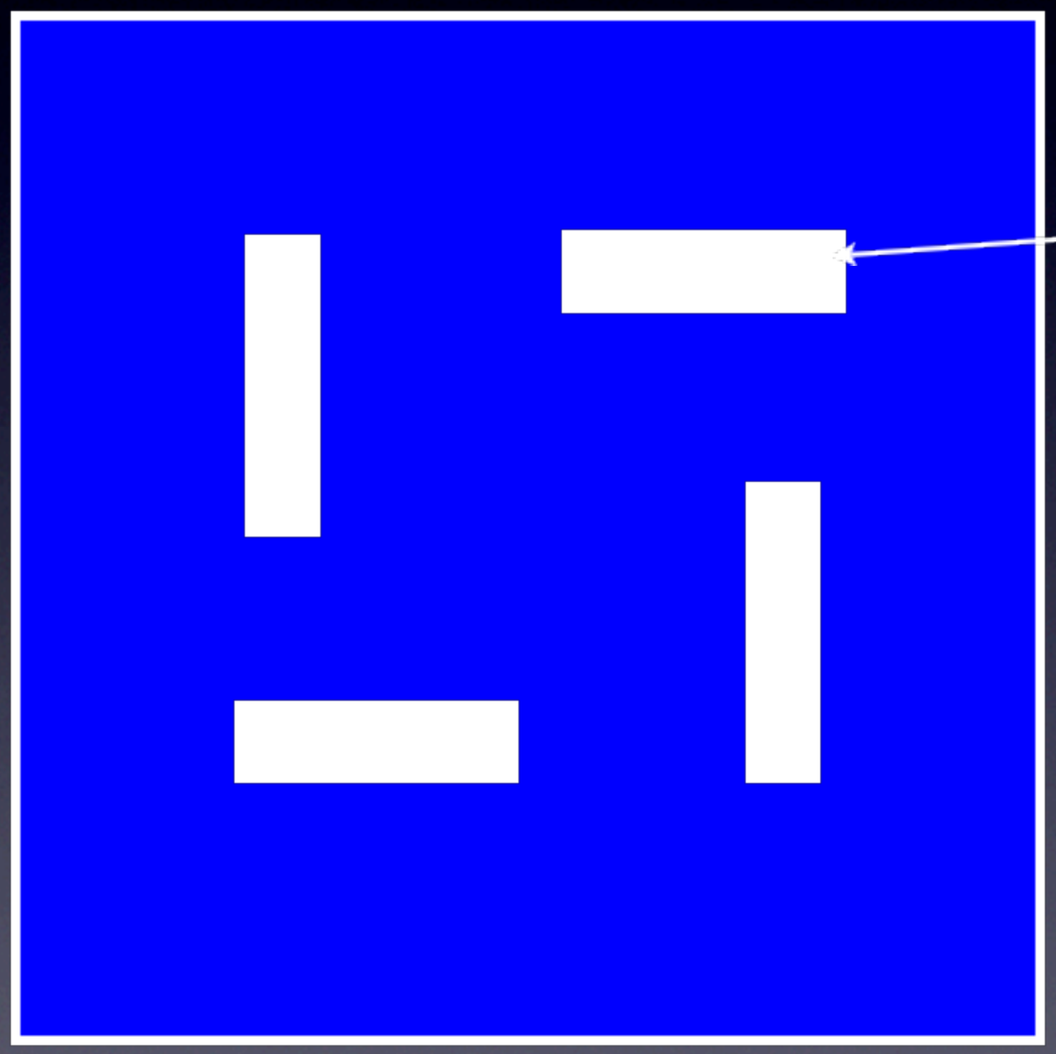


X_L

weight of smallest Z error (mod gauge) = length of defect

Local Architecture

Build a nearly local quantum error correcting architecture:

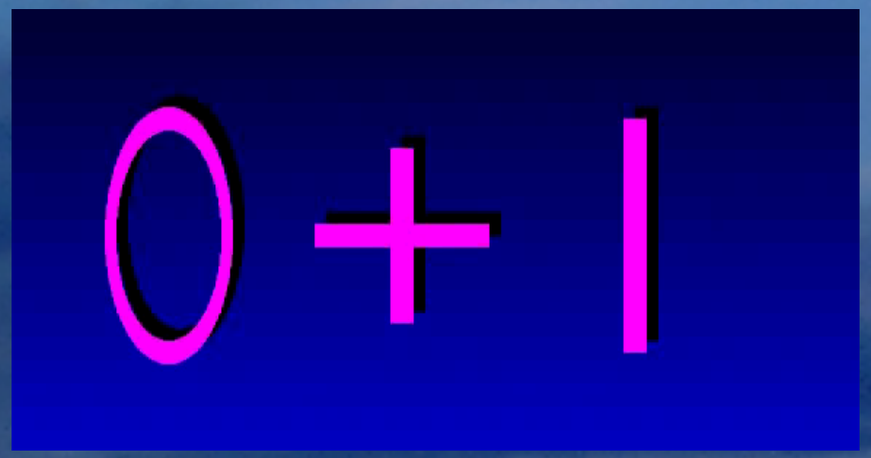


Benefit over toric surface code:
error correction at boundary

Puncturing

Fault-tolerant quantum computing:
physics or computer engineer?

Physical many-body systems with symmetry are a good source of new quantum subsystem codes



Quantum Computing Theory Group

Graduate Students
Gregory Crosswhite (Physics)
Alice Neels (CSE)

Masters Students
Roger Wolfson
(Physics)

Undergraduates
Elizabeth Muhm (CSE/Math)
Alper Sarikaya (CSE/Chem)
William Johnson (CSE)

funded by

ARO NSA NSF